

Electric potential energy, potential difference (voltage) and equipotentials

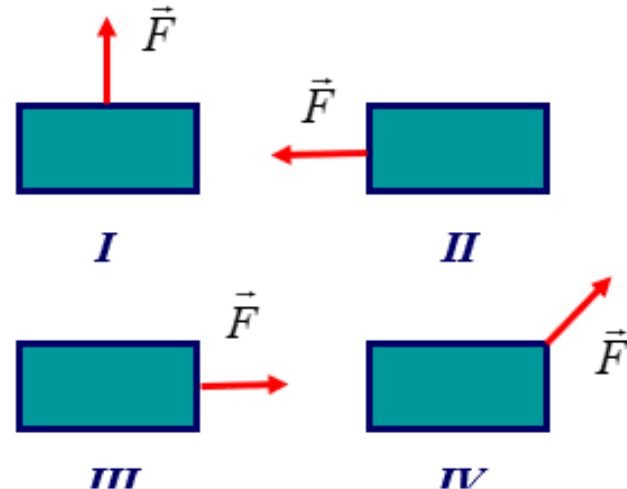
- **uniform electric field**
- **definition of voltage difference**
- **non-uniform field, potential of a point charge**
- **inside a conductor**
- **voltage versus energy (point charge)**
- **equipotentials, corona discharge**
- **applications: potential of a dipole,
potential due to charged rod
Potential due to charged disk or dipole**



Work Done by a Constant Force

1. The right figure shows four situations in which a force is applied to an object. In all four cases, the force has the same magnitude, and the displacement of the object is to the right and of the same magnitude. Rank the situations in order of the work done by the force on the object, from most positive to most negative.

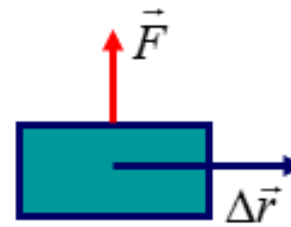
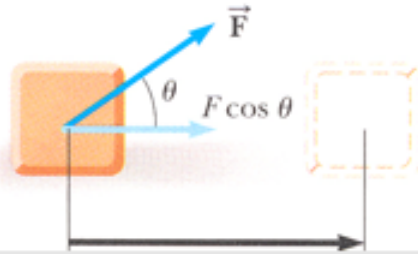
- A. I, IV, III, II
- B. II, I, IV, III
- C. III, II, IV, I
- D. I, IV, II, III
- E. III, IV, I, II



Work Done by a Constant Force

- The work W done a system by an agent exerting a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos\theta$, where θ is the angle between the force and displacement vectors:

$$W \equiv \vec{F} \cdot \Delta\vec{r} = F\Delta r \cos\theta$$



I

$$W_I = 0$$



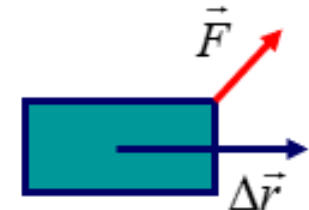
II

$$W_{II} = -F\Delta r$$



III

$$W_{III} = F\Delta r$$



IV

$$W_{IV} = F\Delta r \cos\theta$$

Some Physics I refresher!

Work: General Definition

The general definition of work done by a force must take into account the fact that the force may vary in both magnitude and direction, and that the path followed may also change in direction. All these things can be taken into account by defining work as an integral.

This definition amounts to an infinite sum of the products of the component of force along the path times the path length element

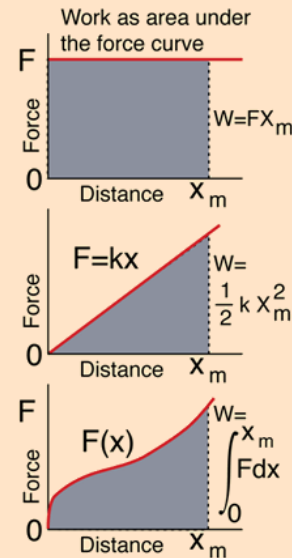
The general idea of work is force times distance moved. This integral evaluates that product in this continuously varying environment.



$$W_{ab} = \int_a^b \vec{F} \cdot d\vec{s}$$

This is taken to represent the work done by the force on the specified path between points a and b. This type of integral is called a line integral because it evaluates the integral of the component of F along a line. The integrand is a scalar product of the vector elements and it can also be written as

$$W_{ab} = \int_a^b F \cos \theta \, ds$$



The basic work relationship $W = Fx$ is a special case which applies only to constant force along a straight line. That relationship gives the area of the rectangle shown, where the force F is plotted as a function of distance. In the more general case of a force which changes with distance, the work may still be calculated as the area under the curve. For example, for the work done to stretch a spring, the area under the curve can be readily determined as the area of the triangle. The power of calculus can also be applied since the integral of the force over the distance range is equal to the area under the force curve:

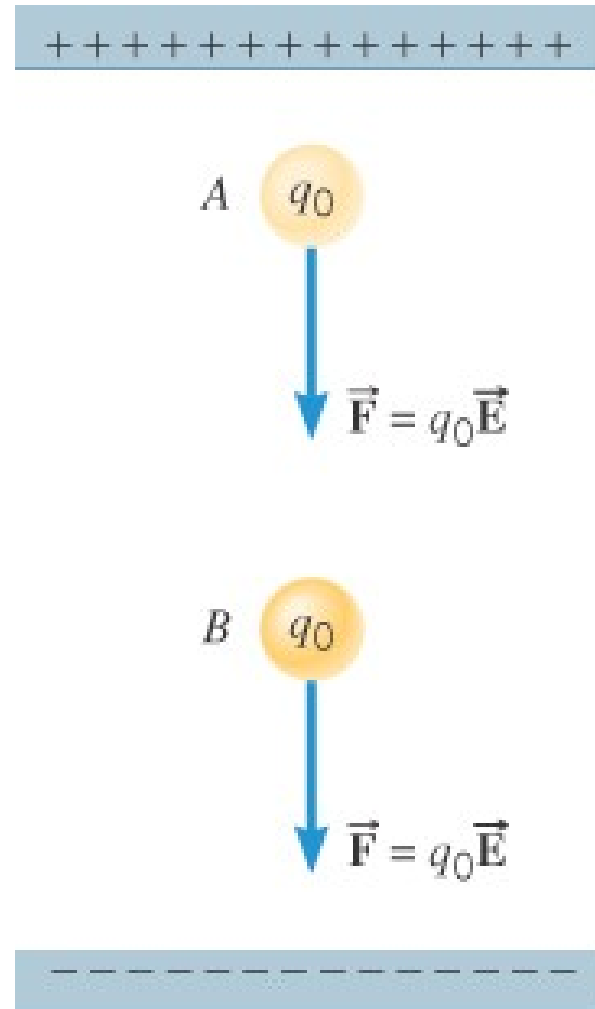
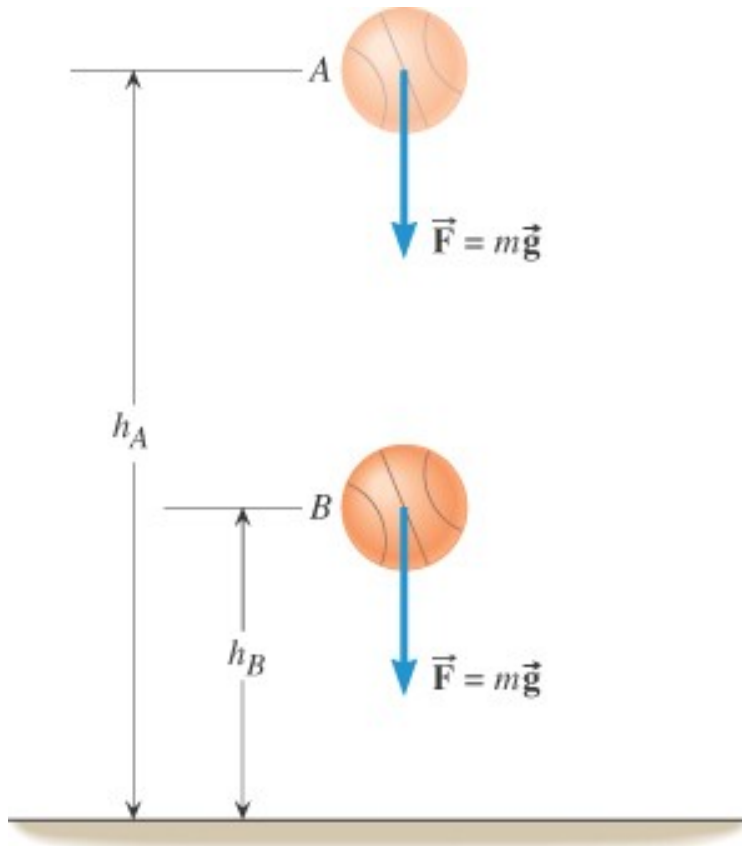
$$\text{Work} = \int_0^{x_m} F(x) \, dx = \int_0^{x_m} kx \, dx = \frac{1}{2} k x_m^2$$

For any function of x , the work may be calculated as the area under the curve by performing the integral

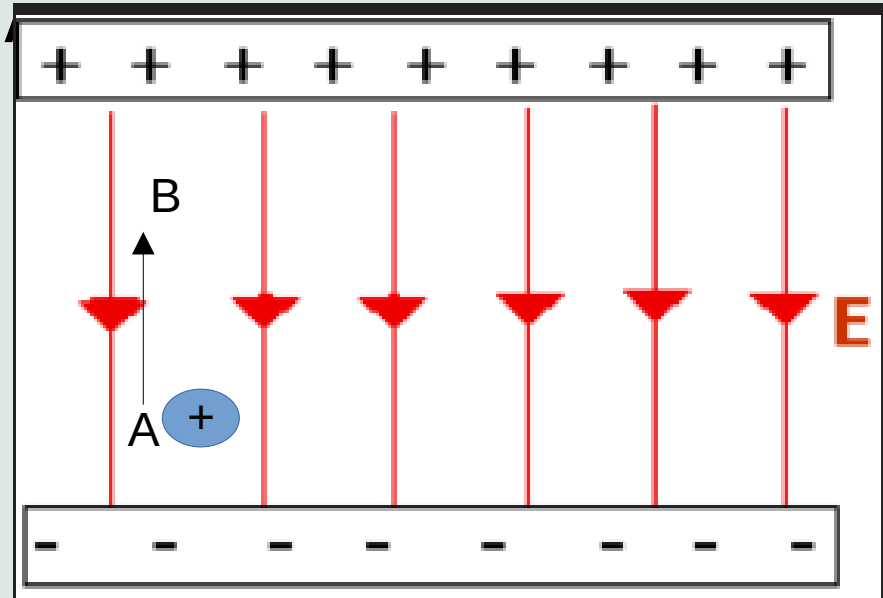
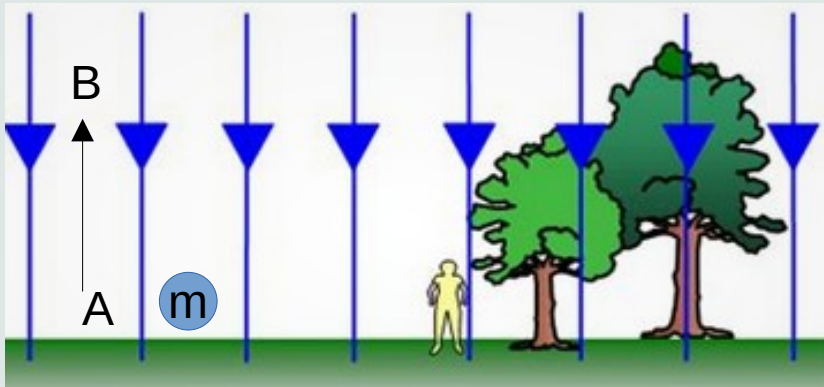
$$\text{Work} = \int_{x_1}^{x_2} F(x) \, dx$$

More general path

ELECTRIC POTENTIAL and POTENTIAL ENERGY



ELECTRIC POTENTIAL



Work done against gravity from A to B = Gravitational Potential Energy of the mass
 $\Delta U = W_{AB} = mg(h_B - h_A)$. Your work is positive. The work done by gravity is negative.
 We could define the change in potential energy per unit mass (for 1 kg) or $g(h_B - h_A)$

g is like E . A field to which a mass or a charge responds.

Work done against the electric field from A to B = change in electric Potential Energy

$$\Delta U = U_B - U_A = -q_0 \int_A^B \vec{E} \cdot d\vec{s}$$

$$\Delta V \equiv \frac{\Delta U}{q_0} = -\int_i^f \vec{E} \cdot d\vec{s}$$

This is the difference
 In potential energy
 Per unit charge = **electric potential**

Unit = volt (J/C) and E (V/m)

We have a new unit for $E \rightarrow V/m$ (its a rate- a gradient – a slope)

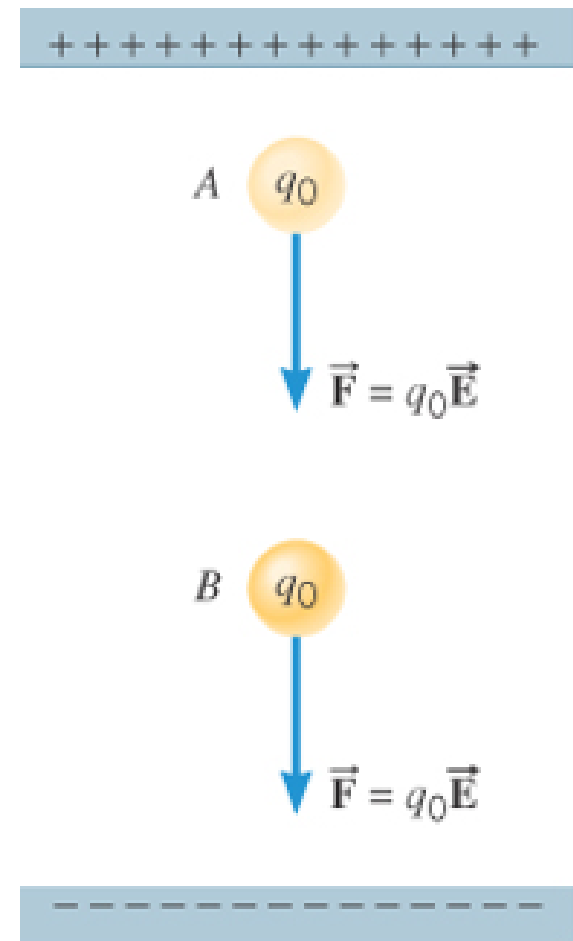
Example 1 Work, Potential Energy, and Electric Potential

The work done by the electric force as the test charge ($+2.0 \times 10^{-6} \text{C}$) moves from A to B is $+5.0 \times 10^{-5} \text{J}$.

- (a) Find the difference in EPE between these points.
- (b) Determine the potential difference between these points.

$$W_{AB} = \text{EPE}_A - \text{EPE}_B$$

$$V_B - V_A = \frac{\text{EPE}_B}{q_o} - \frac{\text{EPE}_A}{q_o} = \frac{-W_{AB}}{q_o}$$

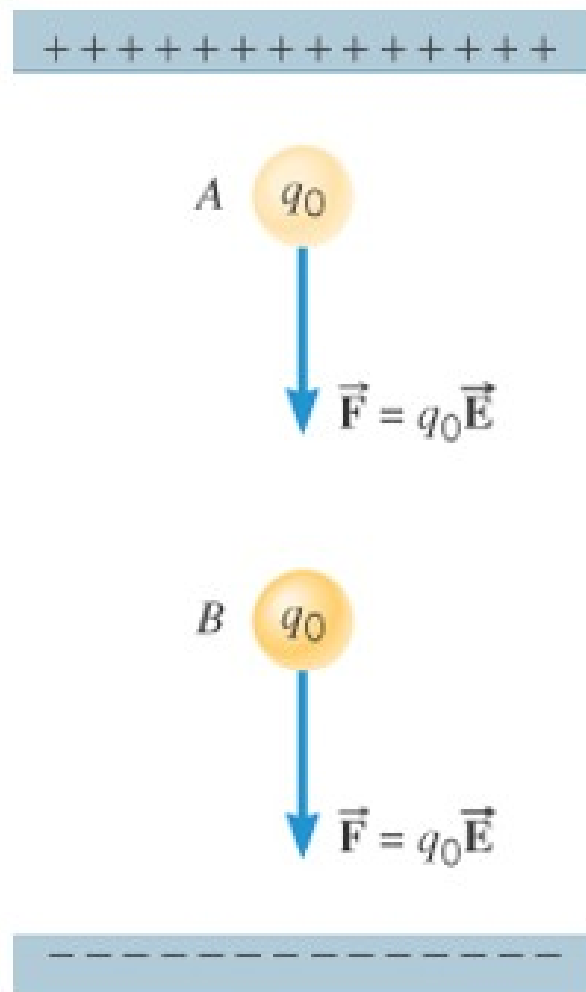


$$(a) \quad W_{AB} = \text{EPE}_A - \text{EPE}_B$$

$$\text{EPE}_B - \text{EPE}_A = -W_{AB} = -5.0 \times 10^{-5} \text{ J}$$

(b)

$$V_B - V_A = \frac{-W_{AB}}{q_o} = \frac{-5.0 \times 10^{-5} \text{ J}}{2.0 \times 10^{-6} \text{ C}} = -25 \text{ V}$$

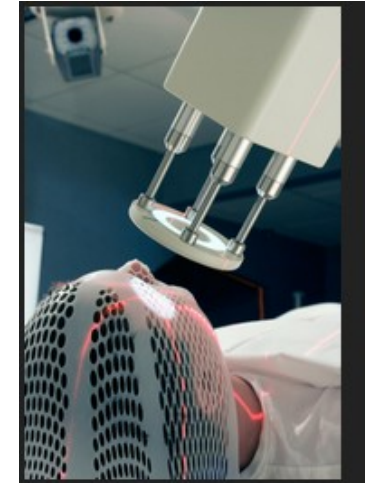
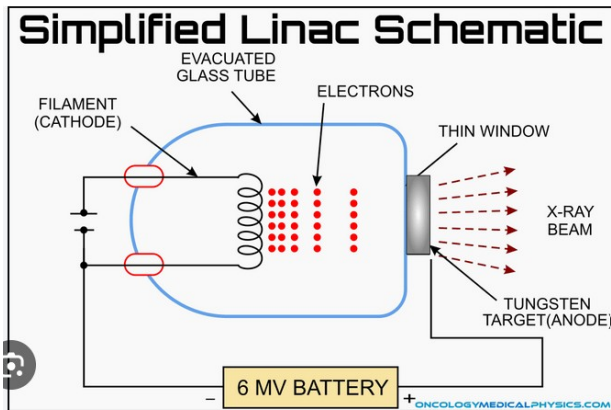


Do textbook 23.3

The eV is the unit of energy.

Textbook application cancer radiotherapy. Using electrons to target superficial Tumors (cms). Like skin cancer.

The voltage is 4 to 20 MV $\rightarrow$ KE $\rightarrow$ destroy tumor
(See Thomas Seyfield – Boston College)



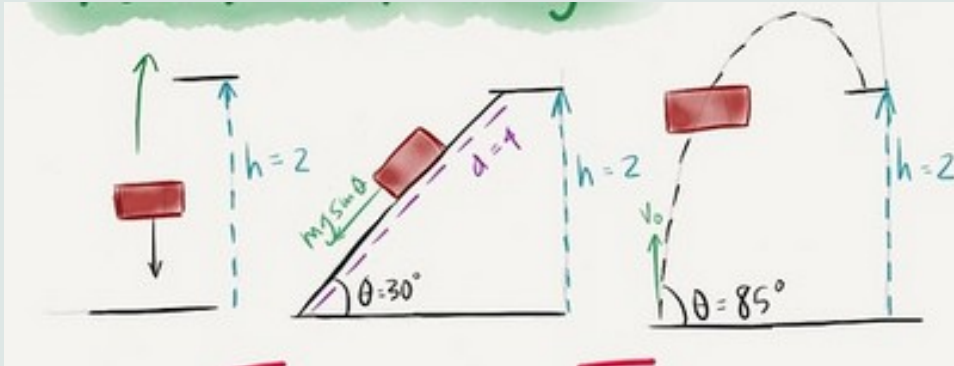
LINACs are multipurpose machines that can produce both high-energy X-rays and electron beams, making them indispensable for various types of cancer therapies.

You also have proton therapy 70 to 250 MeV (million electron volts). Better than X-rays but more expensive

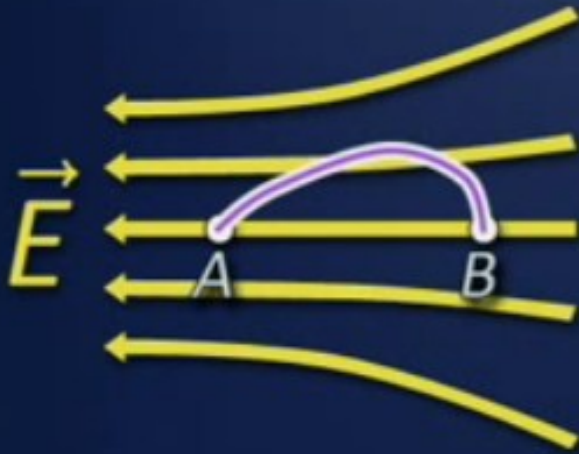
https://www.symmetrymagazine.org/article/december-2008/power-proton-therapy?language_content_entity=und

The voltage is the property of two points.

The electric force is a conservative force because it is derived from a potential energy/
The work done from A to B does not depend on the path.



-Electric force is conservative
-Work is path-independent



- a positive charge moves from high potential to low potential (like mass from high H to low H)
- The work done on a charge is path independent
- voltage is a property of 2 points
- Potential energy depends on a reference level
- Work is done against the electric field

The electric potential difference ΔV_{AB} between two points A and B is the work per unit charge needed to move charge between those points

The voltage is the property of two points.

The electric force is a conservative force because it is derived from a potential energy/
The work done from A to B does not depend on the path.



The voltage is the property of two points $\rightarrow$ 2 terminals.

12V battery means that 1 coulomb of charge is given 12J $\rightarrow$ lose it to the load
12 J / C

Potential energy = charge x (potential difference). $q \Delta V = \Delta U$ (can use eV as energy unit)

voltage

(Electromotive force)

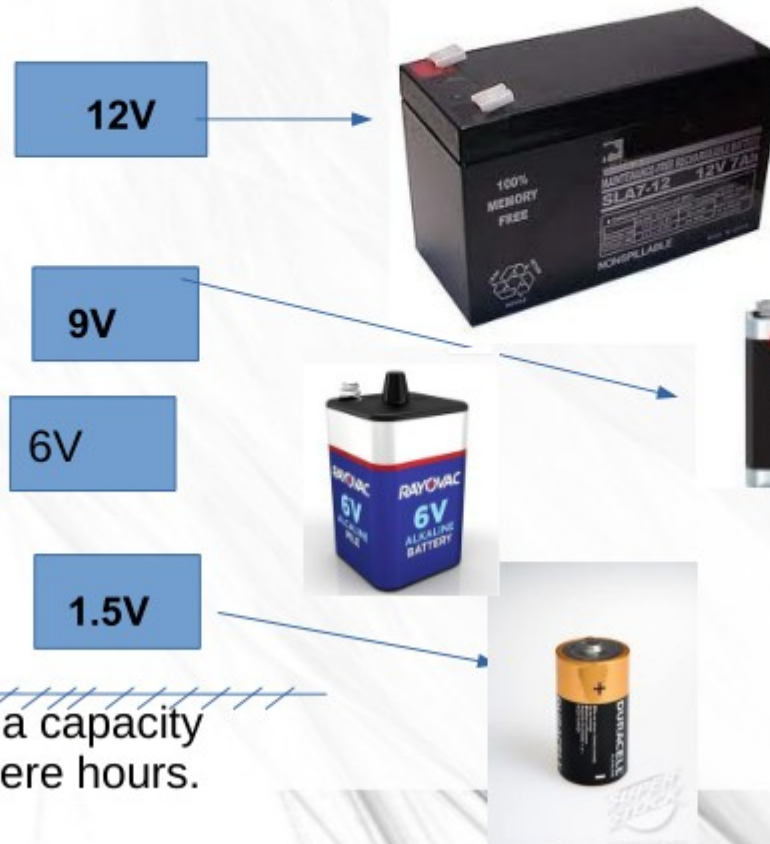
Emf of battery

=Voltage = (potential) energy per unit charge

SI unit is **joules / coulomb = volts (V)** .

Voltage is to a charge what height is to a mass

Increase the
height =
increase the
Energy per
charge



AA size dry cell has a capacity
of about 2 to 3 ampere hours.
3Amp hours →
1Amp in 3 hours

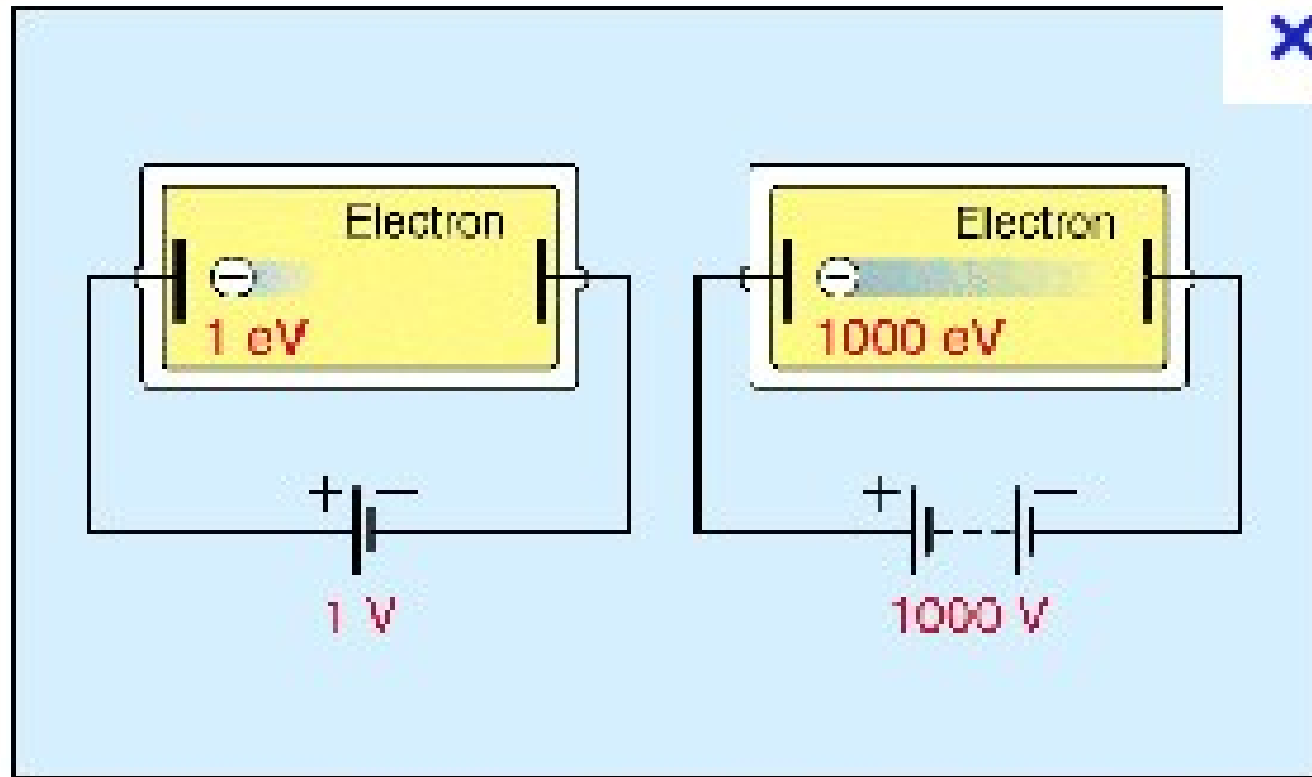
Demo:

Voltmeter
And batteries

One electron volt is the magnitude of the amount by which the potential energy of an electron changes when the electron moves through a potential difference of one volt.

$\Delta U = q\Delta V$ So $\Delta U = e \Delta V \rightarrow$ potential energy of one electron.
.e is the charge of an electron. The unit of energy is eV

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$



- An alpha particle (charge $2e$) moves through a 10-V potential difference. How much work, expressed in eV, is done on the alpha particle?
- A. 5 eV
B. 10 eV
C. 20 eV
D. 40 eV

Example 3 Operating a Headlight

The wattage of the headlight in Figure 19.4 is 60.0 W. Determine the number of particles, each carrying a charge of 1.60×10^{-19} C (the magnitude of the charge on an electron), that pass between the terminals of the 12-V car battery when the headlight burns for one hour.

The Electric Battery

Demo

Volta discovered that electricity could be created if dissimilar metals were connected by a conductive solution called an electrolyte.

This is a simple electric cell.

demo

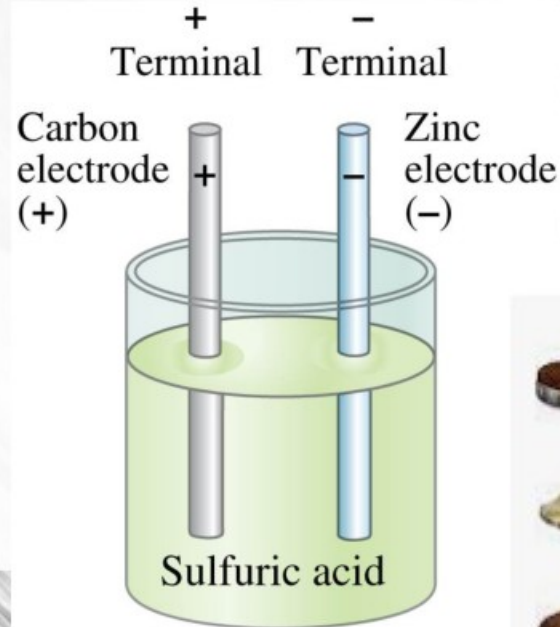


Table 17-1
Some Typical Potential Differences (Voltages)

TABLE 17–1 Some Typical Potential Differences (Voltages)	
Source	Voltage (approx.)
Thundercloud to ground	10^8 V
High-voltage power line	$10^5\text{--}10^6 \text{ V}$
Power supply for TV tube	10^4 V
Automobile ignition	10^4 V
Household outlet	10^2 V
Automobile battery	12 V
Flashlight battery	1.5 V
Resting potential across nerve membrane	10^{-1} V
Potential changes on skin (EKG and EEG)	10^{-4} V



The voltage is the property of 2 points.
It's the difference in potential.

If there are no 2 points, charges can not flow.

Why is this person not electrocuted?

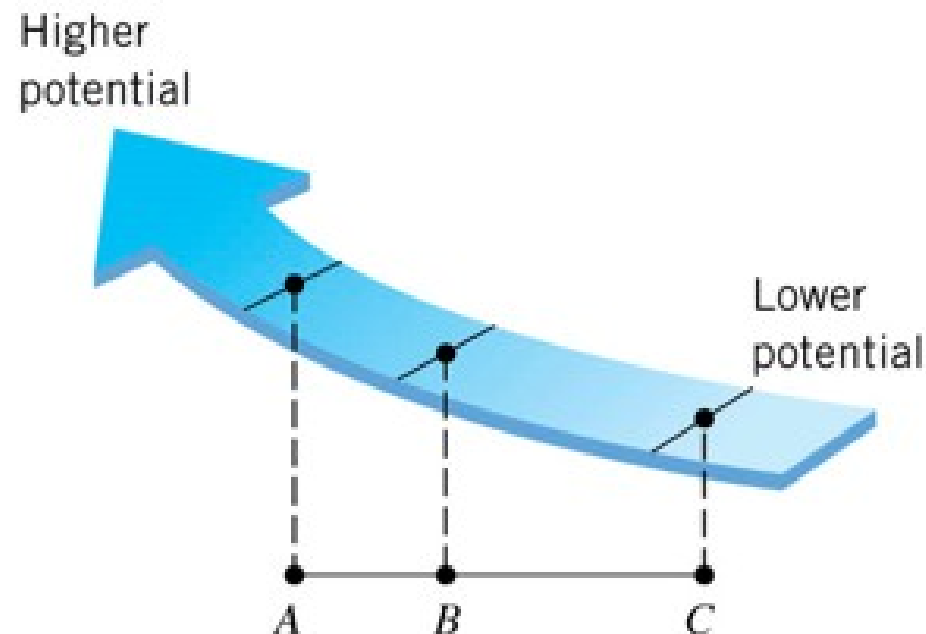
His body is on an equipotential .



Conceptual Example 2 The Accelerations of Positive and Negative Charges

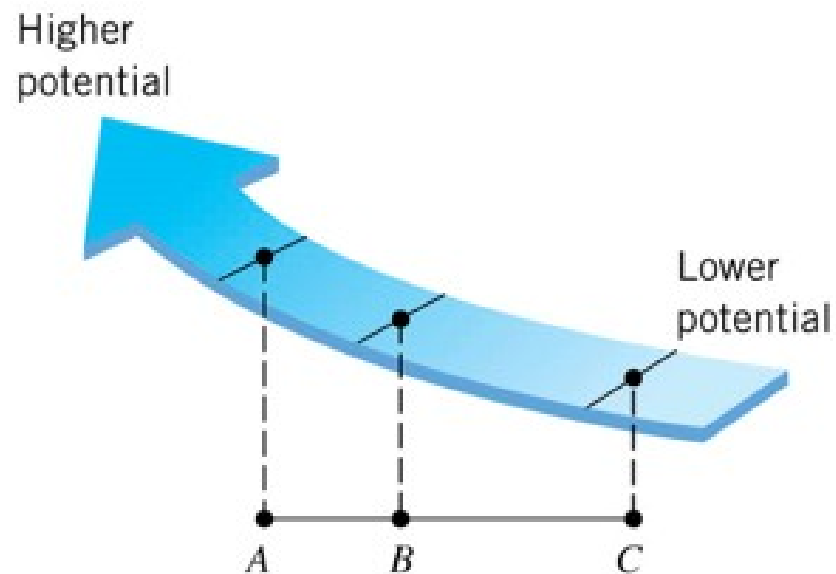
A positive test charge is released from A and accelerates towards B. Upon reaching B, the test charge continues to accelerate toward C. Assuming that only motion along the line is possible, what will a negative test charge do when released from rest at B?

**A positive charge always
Moves from high potential to low potential
A negative charge always
Moves from low potential to high potential**



A positive charge accelerates from a region of higher electric potential toward a region of lower electric potential.

A negative charge accelerates from a region of lower potential toward a region of higher potential.



Thunderstorm Voltage

Fair weather electric field: $\sim 100 \text{ V/m}$

Breakdown electric field: $\sim 3 \text{ MV/m}$

Under thunderstorm: $\sim 10,000 \text{ V/m}$

Altitude, cloud bottom: 2 km

Potential difference between
cloud and ground?



The calculus part

- Start $dW = \vec{F} \cdot d\vec{s}$
- Then $dW = q_0 \vec{E} \cdot d\vec{s}$
- So $W = q_0 \int_i^f \vec{E} \cdot d\vec{s}$

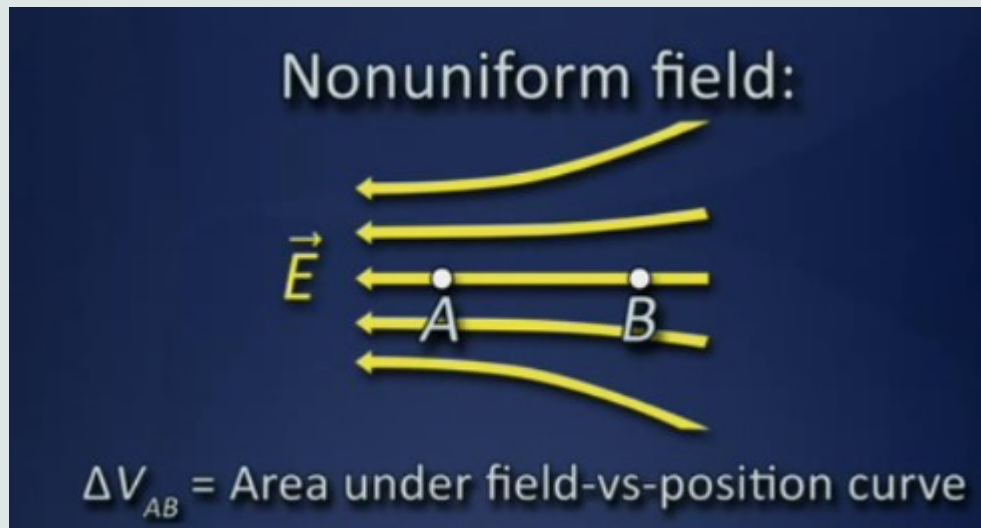
$$\Delta U = U_f - U_i = -W = -q_0 \int_i^f \vec{E} \cdot d\vec{s}$$

The electric potential $V = \frac{U}{q}$

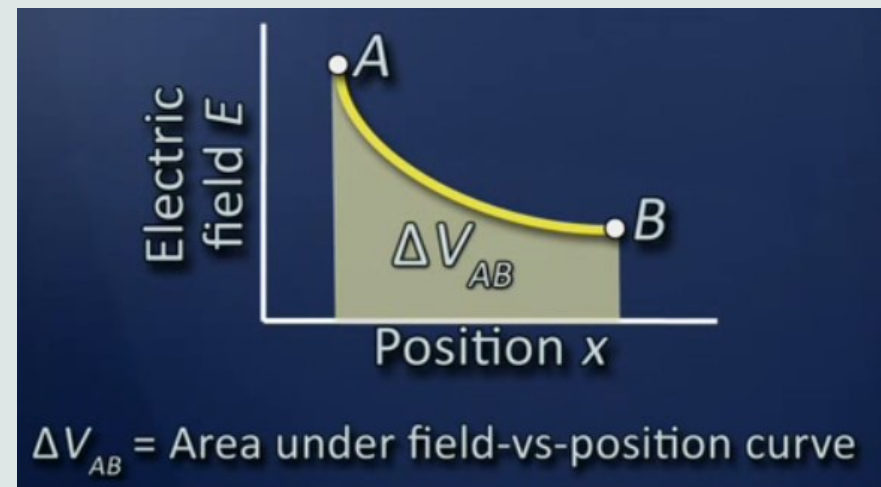
$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}$$

$$\Delta V \equiv \frac{\Delta U}{q_0} = - \int_i^f \vec{E} \cdot d\vec{s}$$

Non Uniform field:



The work done per unit charge $\rightarrow$
Integral. Electric weakens out.
Easier at the end.



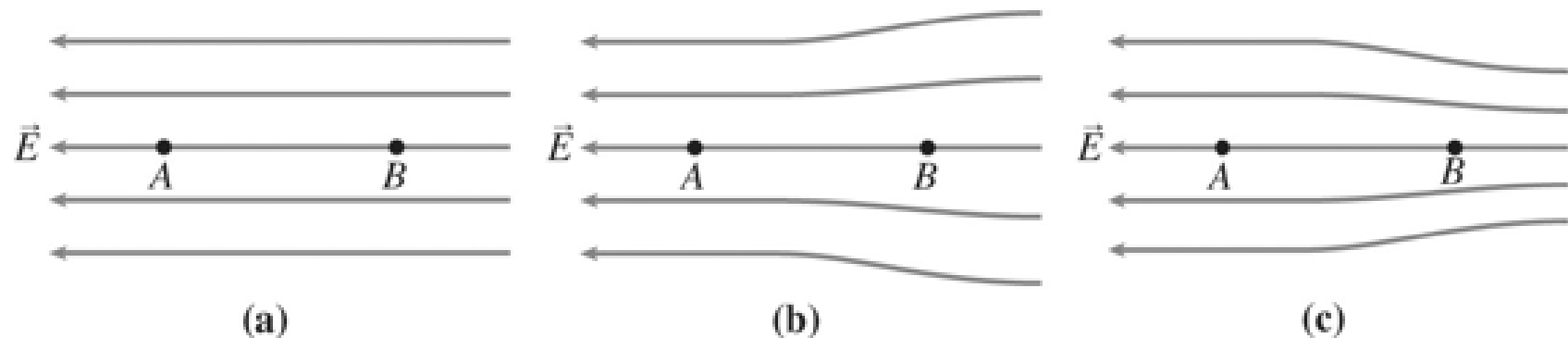
- The figure shows three straight paths AB of the same length, each in a different electric field. Which one of the three has the largest potential difference between the two points?

A. (a)

B. (b)

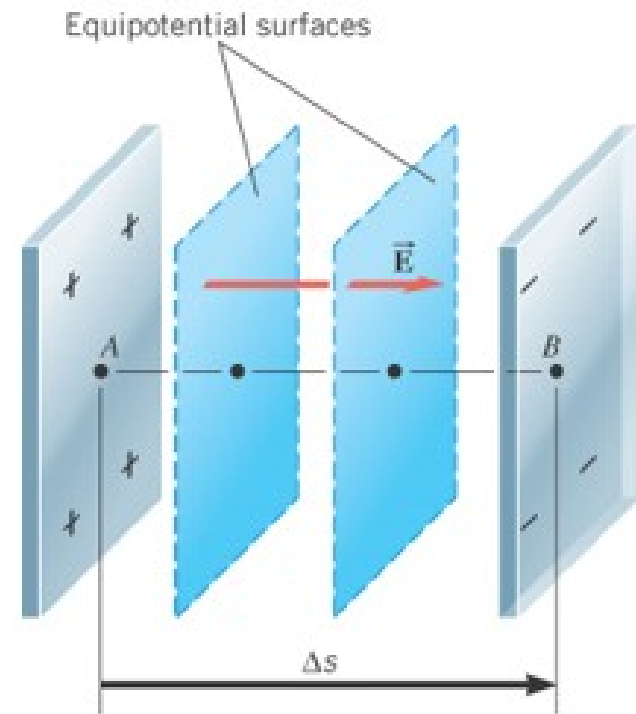
C. (c)

Hint: make a graph E vs x and shade the area below
The graph in each case.

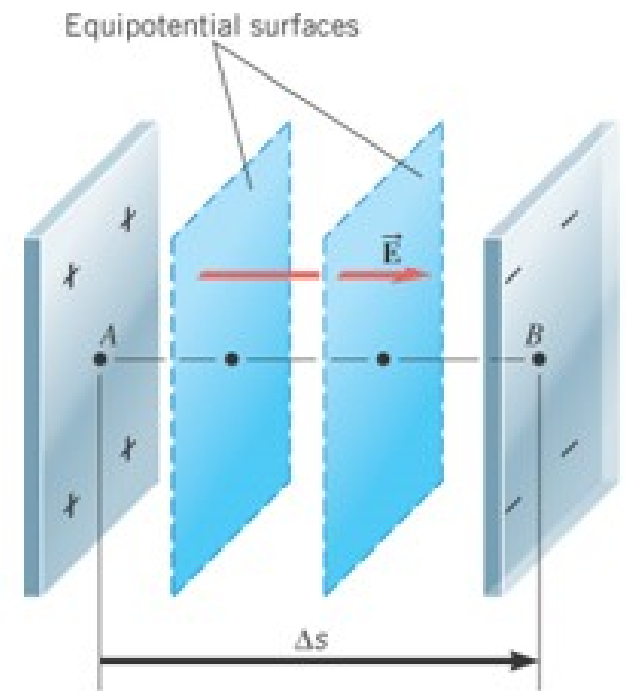


Example 9 The Electric Field and Potential Are Related

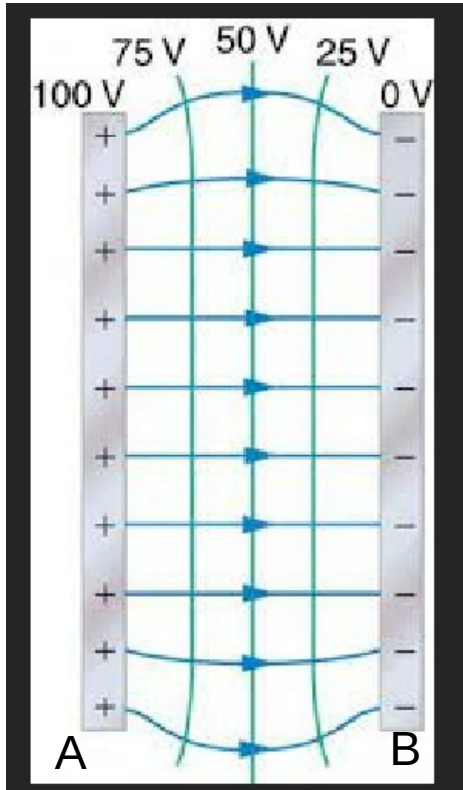
The plates of the capacitor are separated by a distance of 0.032 m, and the potential difference between them is $V_B - V_A = -64\text{V}$. Between the two equipotential surfaces shown in color, there is a potential difference of -3.0V . Find the spacing between the two colored surfaces.



$$E = -\frac{\Delta V}{\Delta s} = \frac{-64 \text{ V}}{0.032 \text{ m}} = 2.0 \times 10^3 \text{ V/m}$$



$$\Delta s = -\frac{\Delta V}{E} = -\frac{-3.0 \text{ V}}{2.0 \times 10^3 \text{ V/m}} = 1.5 \times 10^{-3} \text{ m}$$



What is E if the distance between the plates is 1mm?

Trace V as a function of distance

What is $V_A - V_B$? What is $V_B - V_A$

From your book ->

Example 23.3 Electric force and electric potential

WITH VARIATION PROBLEMS

A proton (charge $+e = 1.602 \times 10^{-19} \text{ C}$) moves a distance $d = 0.50 \text{ m}$ in a straight line between points a and b in a linear accelerator. The electric field is uniform along this line, with magnitude

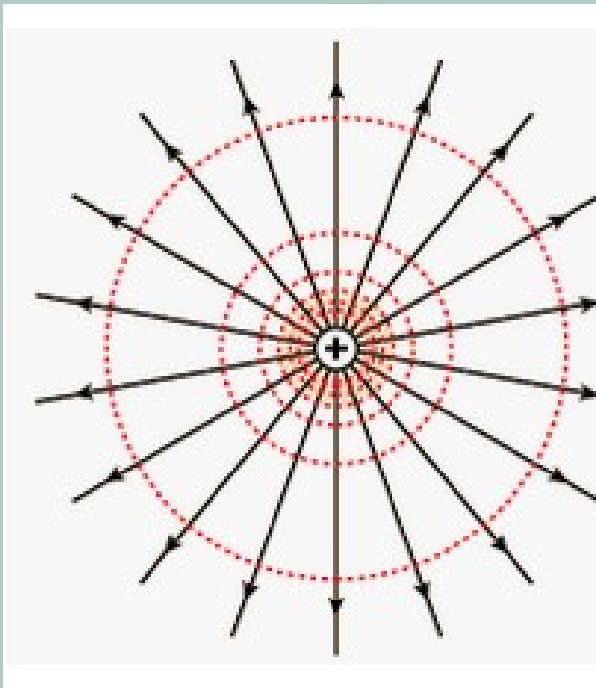
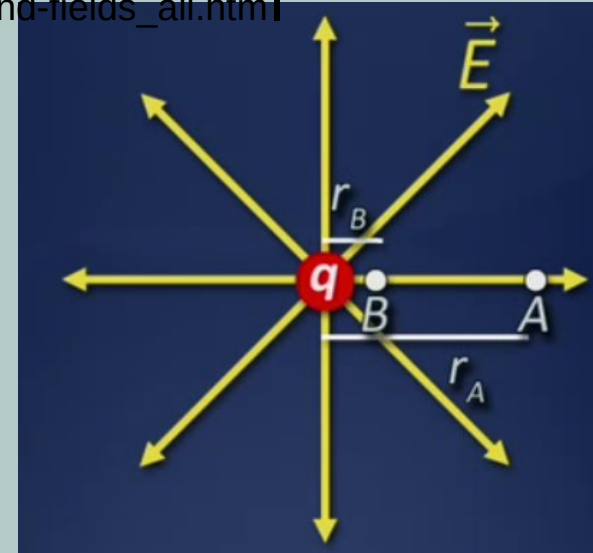
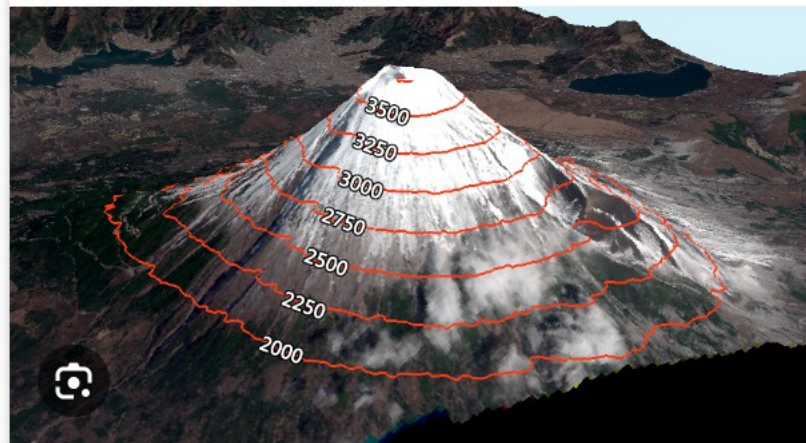
$$E = 1.5 \times 10^7 \text{ V/m} = 1.5 \times 10^7 \text{ N/C} \text{ in the direction from } a \text{ to } b.$$

Determine (a) the force on the proton; (b) the work done on it by the field; (c) the potential difference $V_a - V_b$.

Potential of a point charge

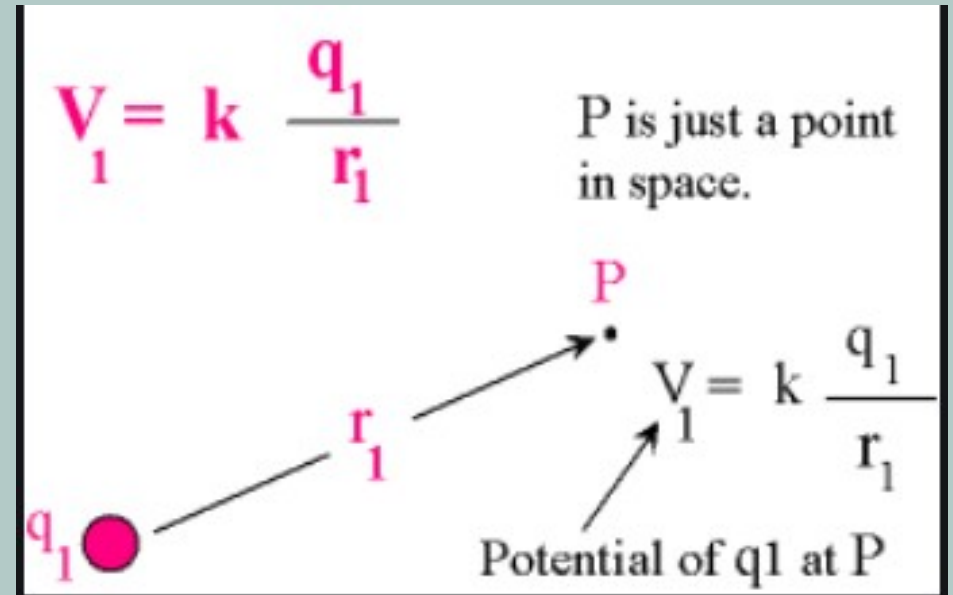
Show it using calculus - Show the java simulation - > show V is inversely proportional to r and proportional to Q
Also we supposed $V = 0$ for $r = \text{infinity}$. Voltage is a property of 2 points.

https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_all.html



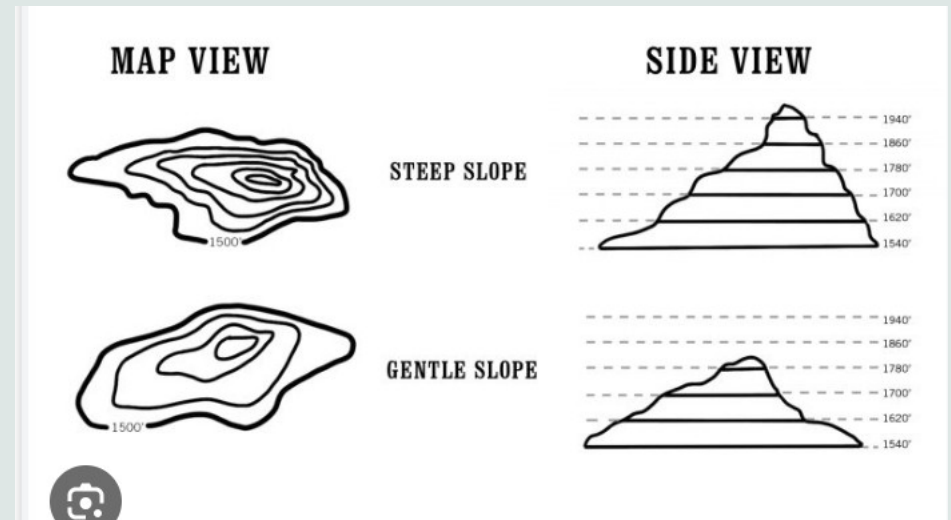
$$V_1 = k \frac{q_1}{r_1}$$

P is just a point in space.

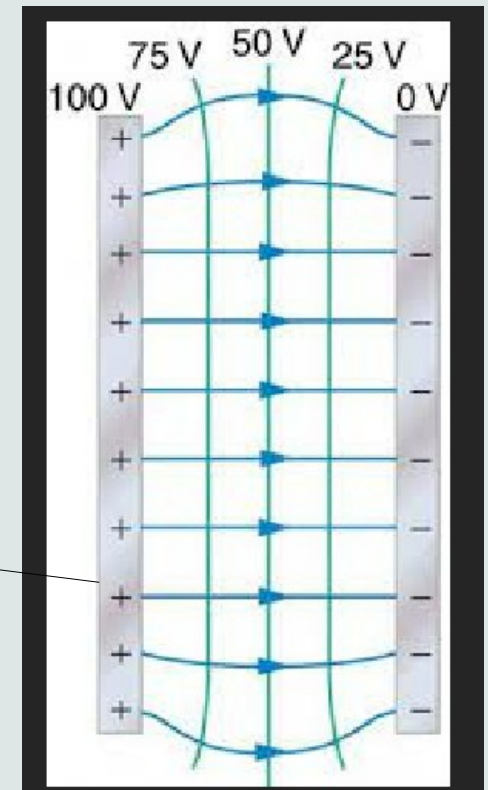
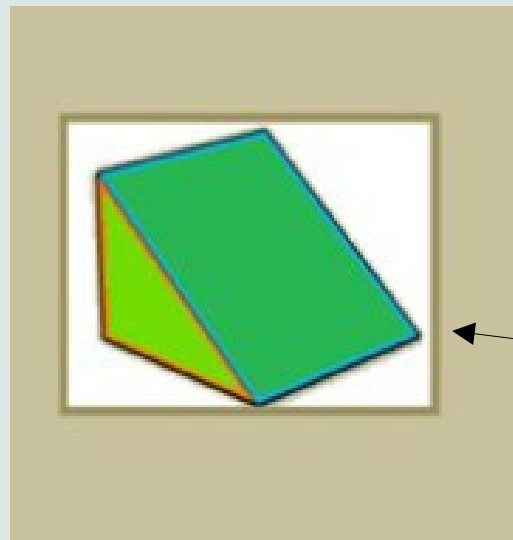
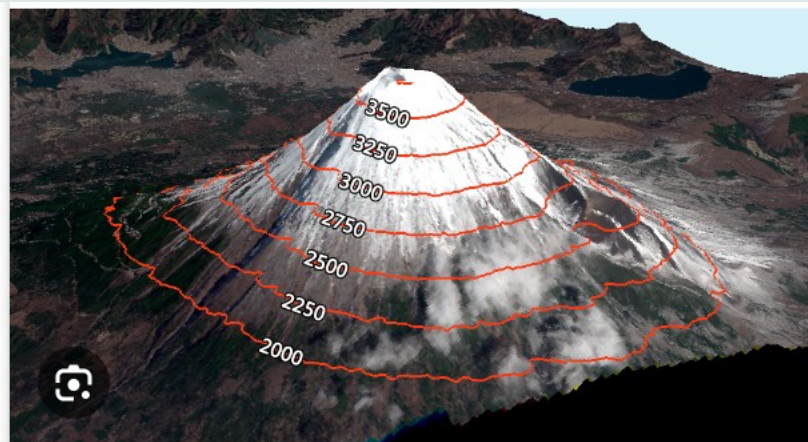

$$V_1 = k \frac{q_1}{r_1}$$

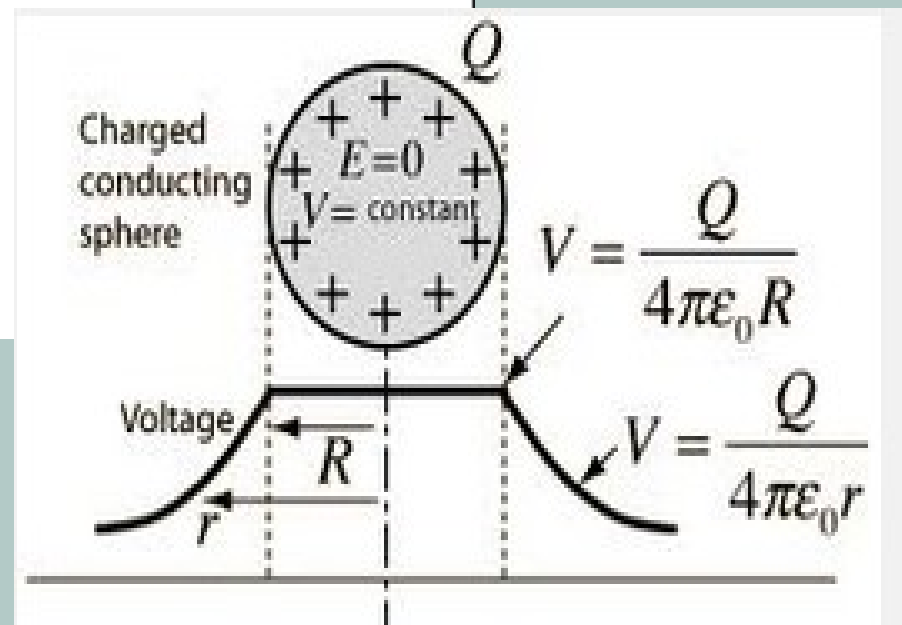
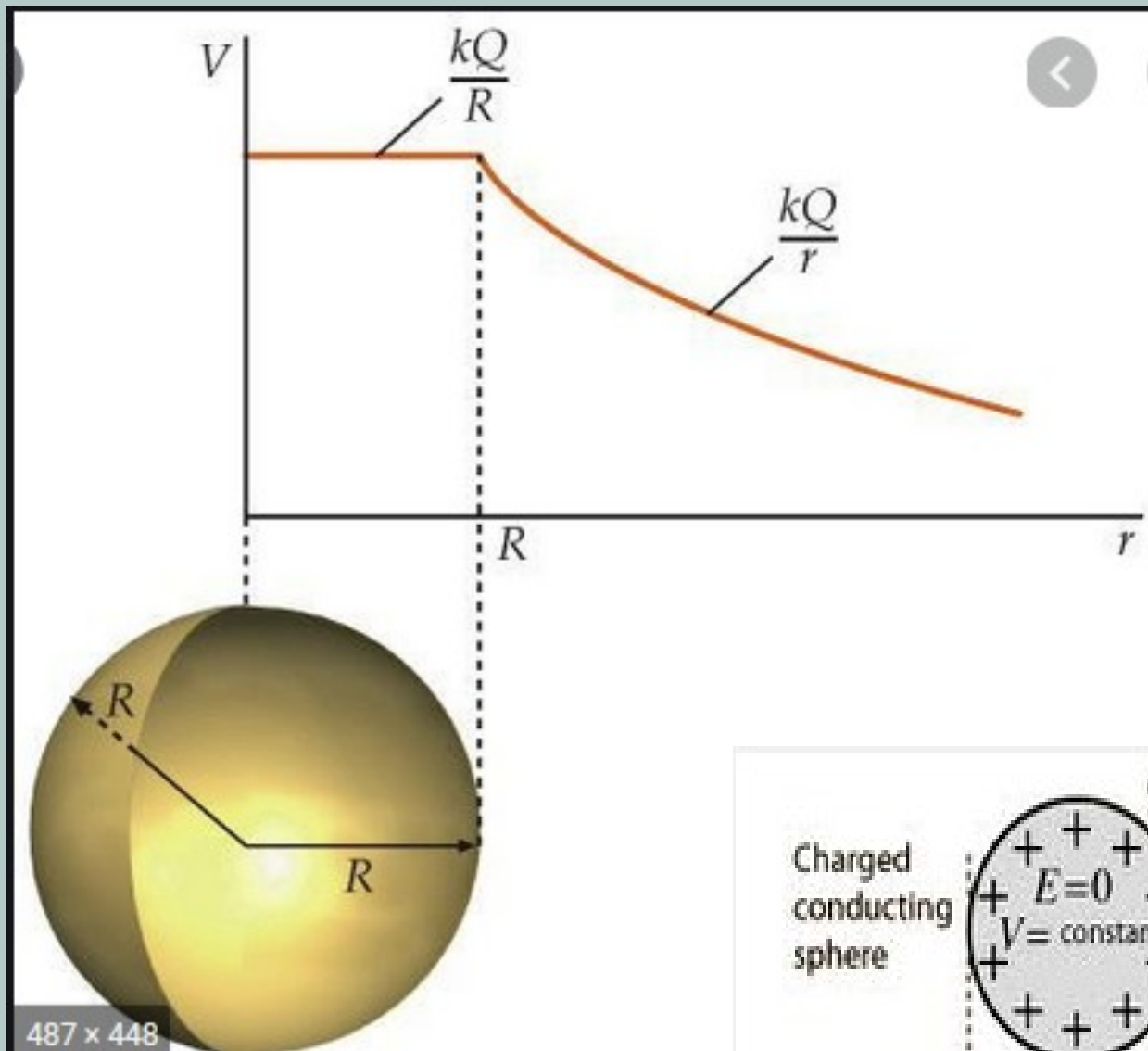
Potential of q_1 at P

equipotentials



No work is done along the contour lines. Gravity (component) is perpendicular to the lines. If a charge is moved along the Contour lines, there is no work done. Like walking at the same height.





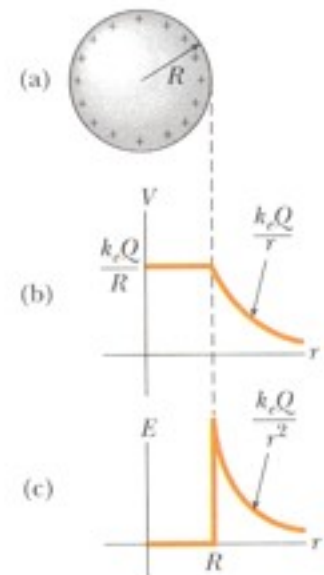
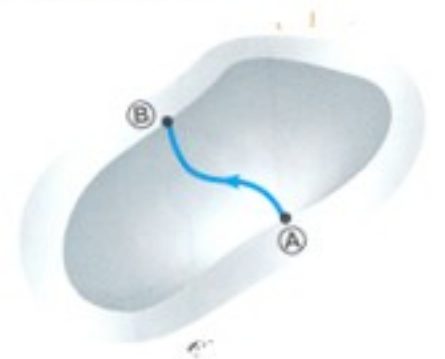
Potential Due to a Charged Isolated Conductor

- According to Gauss' law, the charge resides on the conductor's outer surface.
- Furthermore, the electric field just outside the conductor is perpendicular to the surface and field inside is zero.

- Since

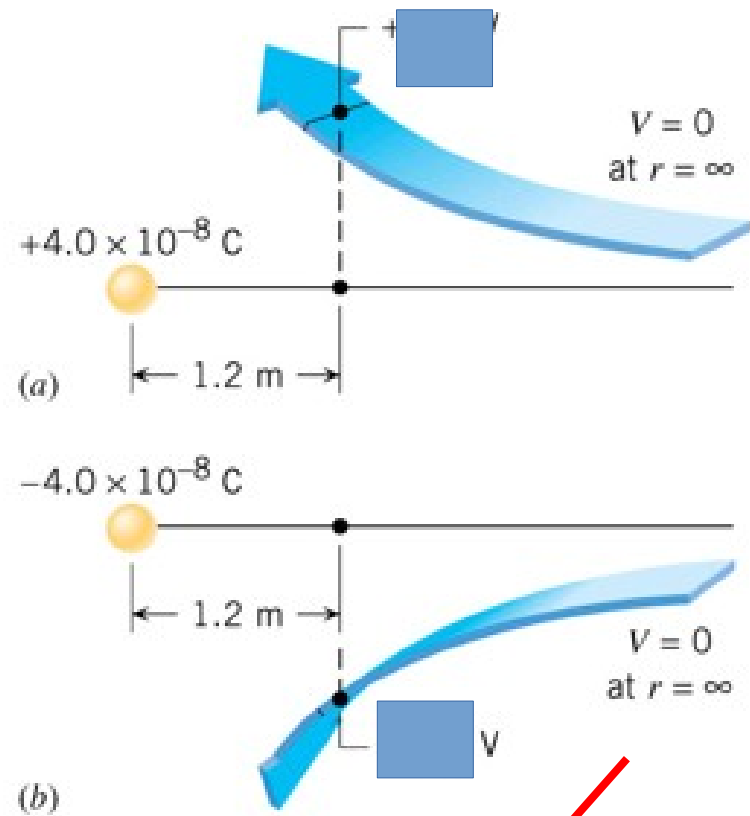
$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{S} = 0$$

- Every point on the surface of a charged conductor in equilibrium is at the same electric potential.
- Furthermore, the electric potential is constant everywhere inside the conductor and equal to its value to its value at the surface.



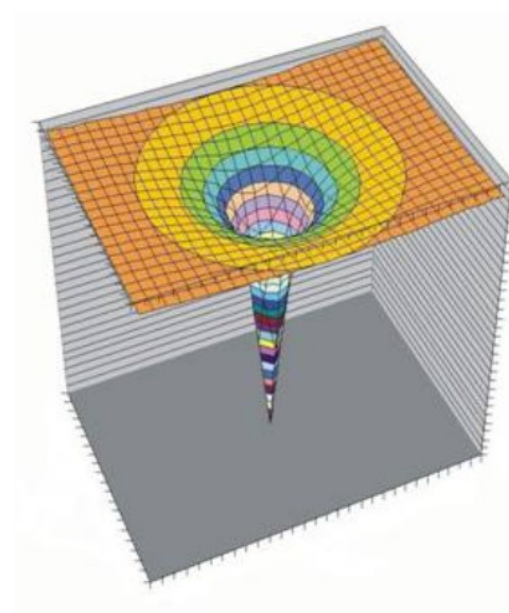
Example 5 The Potential of a Point Charge

Using a zero reference potential at infinity, determine the amount by which a point charge of $4.0 \times 10^{-8} \text{ C}$ alters the electric potential at a spot 1.2 m away when the charge is (a) positive and (b) negative.



What will happen if you place a positive Charge at 1.2 m from the source
In each case?

What about a negative charge.
(climbs up the potential.
Note that a negative mass does not exist).

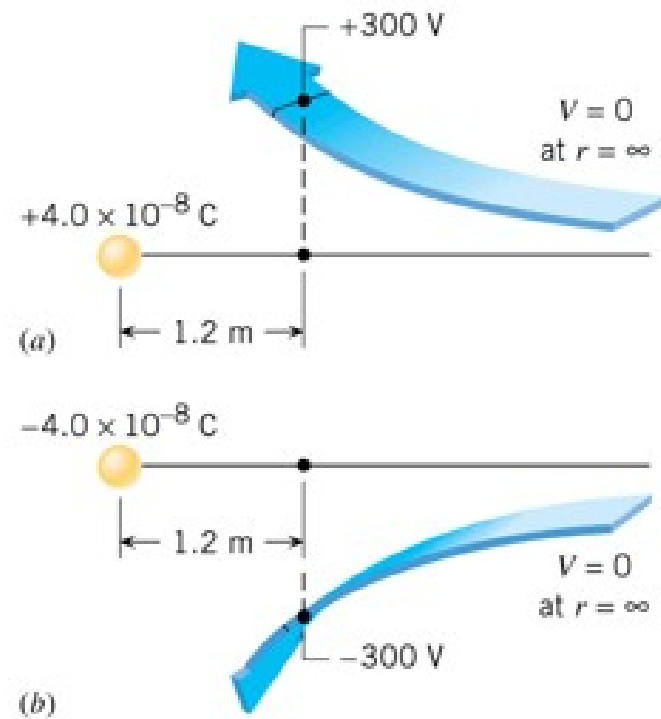


(a)

$$V = \frac{kq}{r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(+4.0 \times 10^{-8} \text{ C})}{1.2 \text{ m}} = +300 \text{ V}$$

(b)

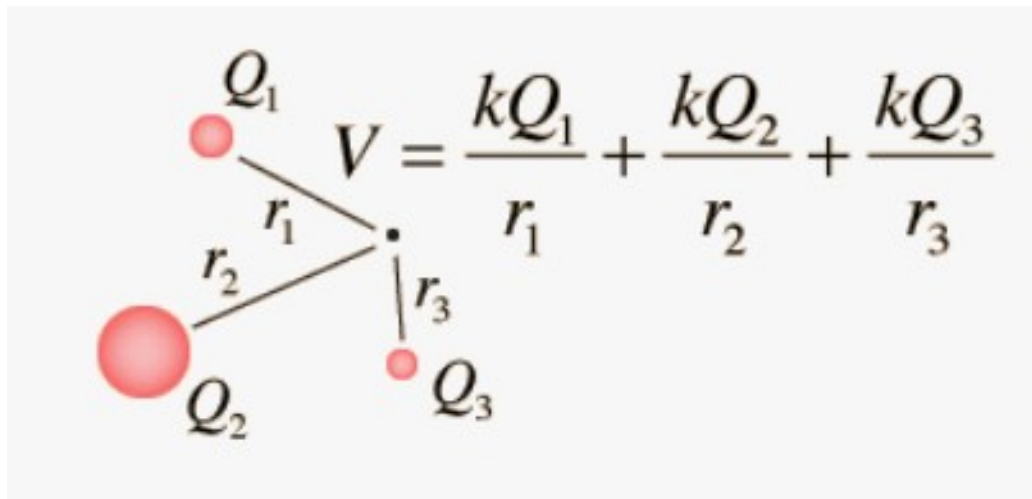
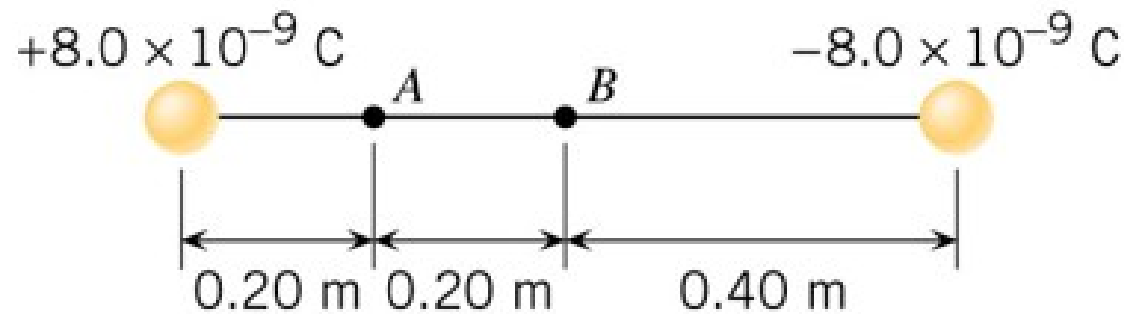
$$V = -300 \text{ V}$$

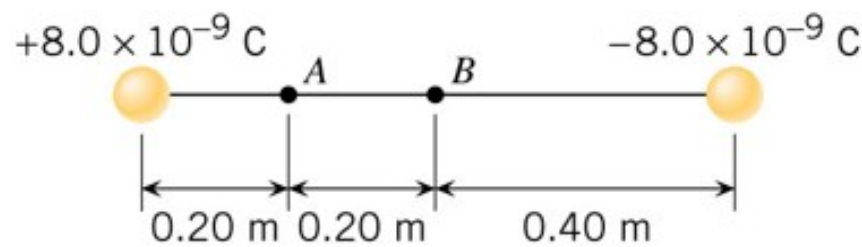


Example 6 The Total Electric Potential

At locations A and B, find the total electric potential.

Voltage from several point charges → Use the superposition principle →
Add the voltages from each charge,

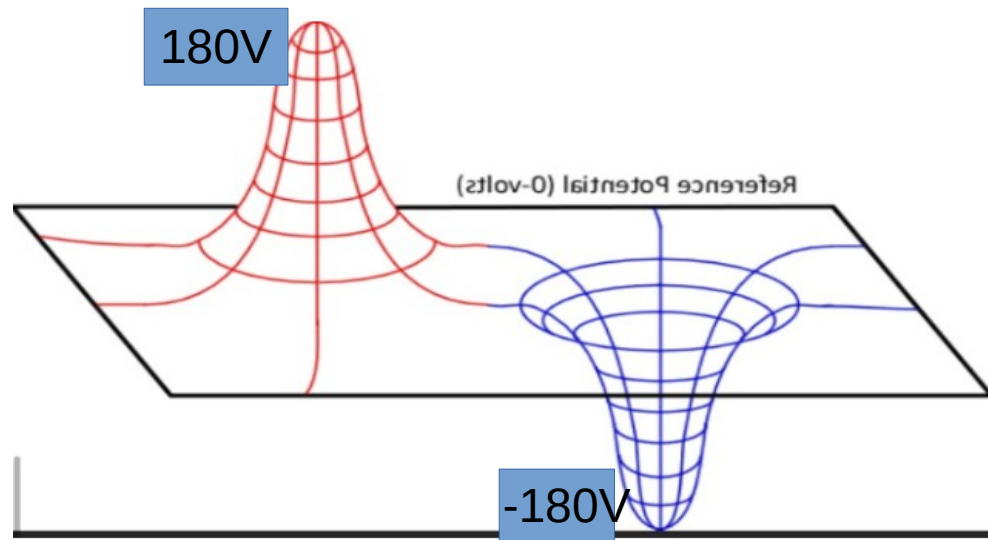




$$V_A = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(+8.0 \times 10^{-8} \text{ C})}{0.20 \text{ m}} + \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(-8.0 \times 10^{-8} \text{ C})}{0.60 \text{ m}} = +240 \text{ V}$$

This is for $V=0$ at infinity.
Define the reference level first

$$V_B = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(+8.0 \times 10^{-8} \text{ C})}{0.40 \text{ m}} + \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(-8.0 \times 10^{-8} \text{ C})}{0.40 \text{ m}} = 0 \text{ V}$$

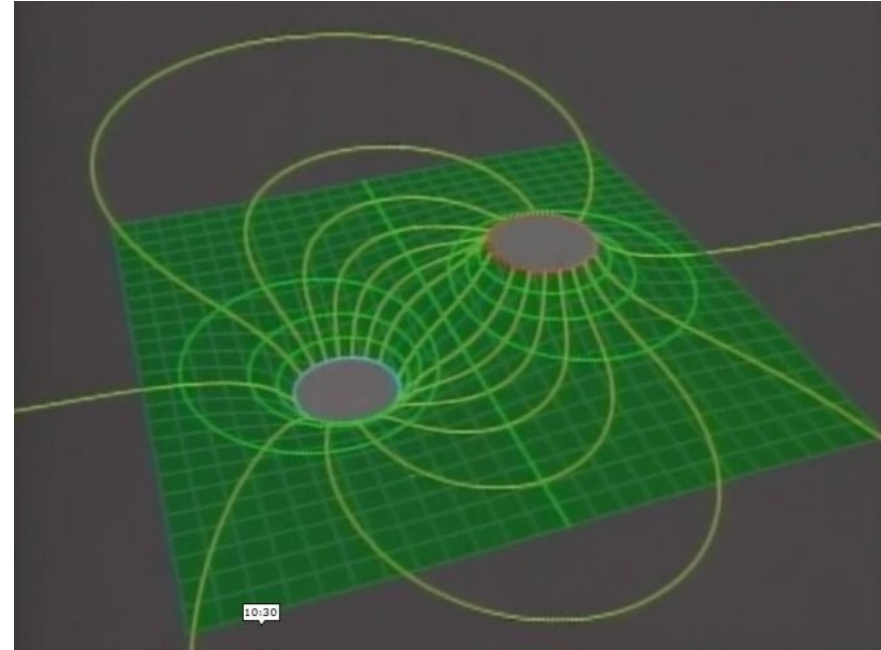
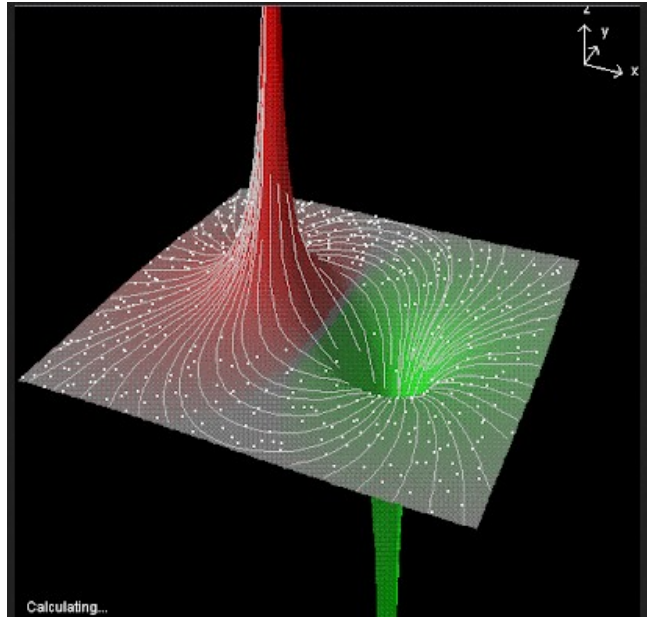


If you take $V=0$ between the charges (B)
Then $V=180\text{V}$ (Q) and $V = -180\text{V}$ (- Q)
So the voltage between the charges
Is 250V

In a house $\rightarrow$ -120V AC and 120VAC (2 hot lines)
So we can get 240V AC (laundry)

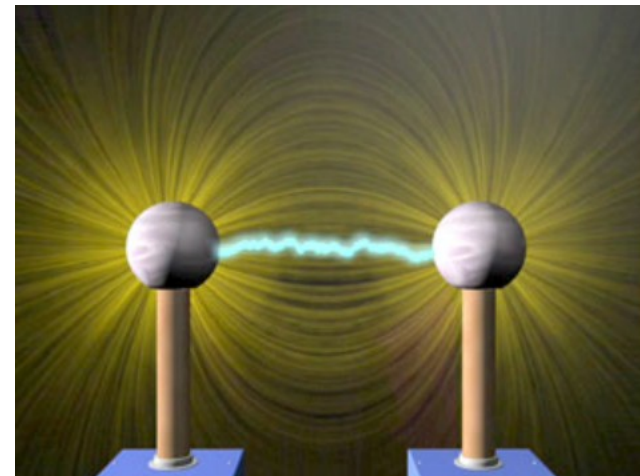
Simulation. Show that with a dipole, there is a point, midpoint where the Potential is 0.

Building on previous slide. What the potential energy of a charge 4nC placed at A
What about -4nC ? Potential energy is supposed to be 0 at infinity
Negative U means its binding.



If $E > 3\text{MV/m} \rightarrow$ discharge

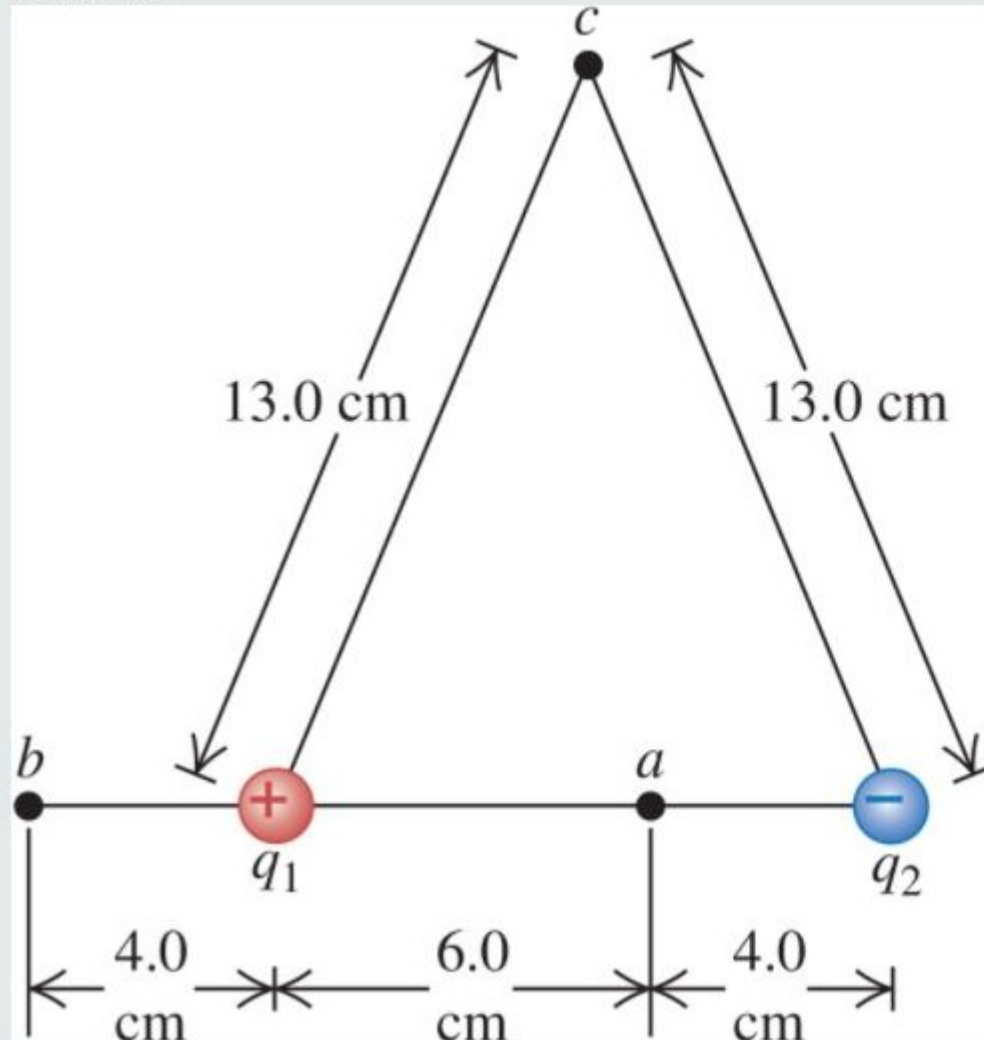
@ right
Note that the charges are not point like
But spheres. Inside, the voltage
Does no change and is constant.
 $V_{\text{inside}} = V_{\text{surface}}$.
 $E = 0$ inside.



Do textbook 23.4

An electric dipole consists of point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ placed 10.0 cm apart (Fig. 23.13). Compute the electric potentials at points a , b , and c .

FIGURE 23.13



This is from your textbook
Add voltages

$$V_c = 0$$

$$V_a = -900 \text{ V}$$

$$V_b = 1929 \text{ V}$$

Dipole again. You see that finding the voltage at a point P is easier than finding E
 Because V is a scalar. We just add numbers. (one is negative).
 You see that $r \gg d$ then V decreases as the square of the distance.

$$V = kq \left[\frac{1}{r_+} - \frac{1}{r_-} \right] = kq \left[\frac{r_- - r_+}{r_+ r_-} \right]$$

For cases where $r \gg d$, this can be approximated by

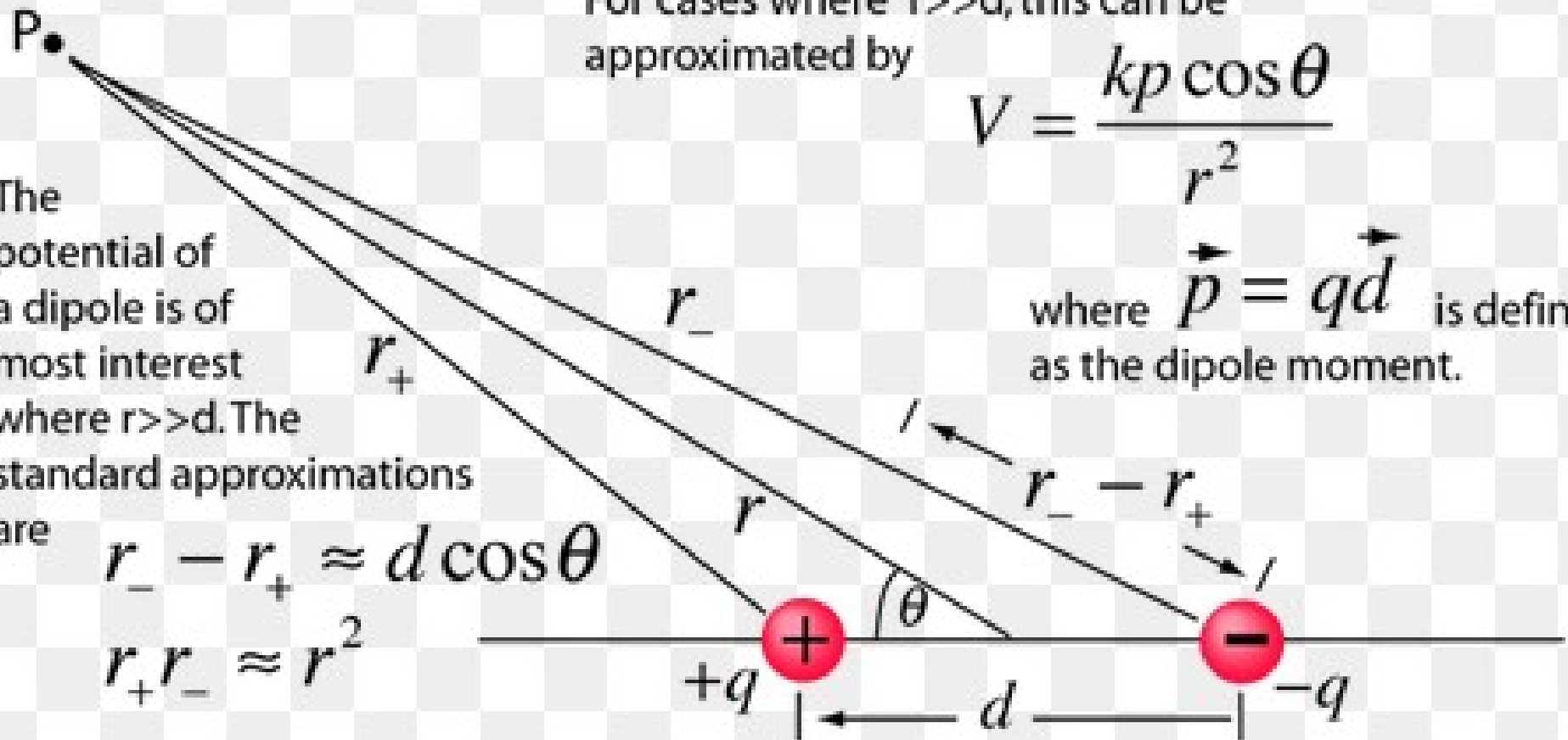
$$V = \frac{kp \cos \theta}{r^2}$$

The potential of a dipole is of most interest where $r \gg d$. The standard approximations are

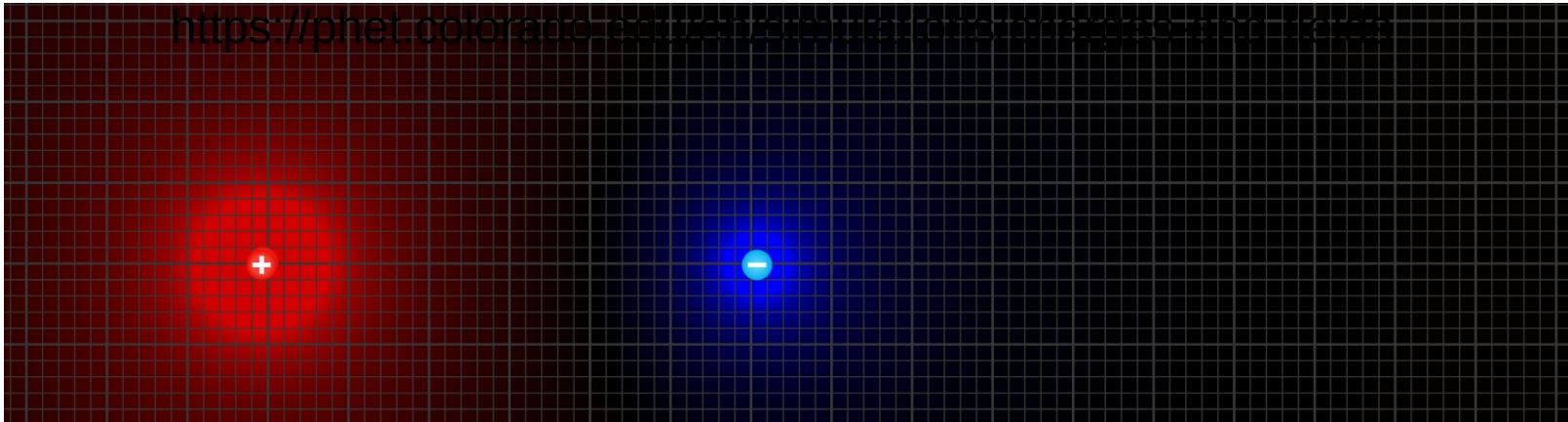
$$r_- - r_+ \approx d \cos \theta$$

$$r_+ r_- \approx r^2$$

where $\vec{p} = q\vec{d}$ is defined as the dipole moment.



<https://phet.colorado.edu/en/simulations/charges-and-fields>

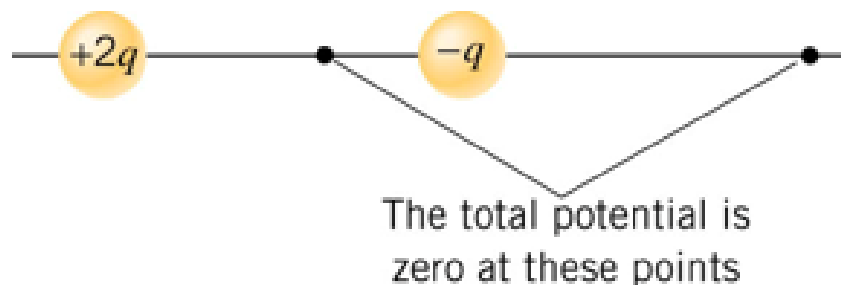


First Place 2 positive charge (red) and one negative charge (blue) 6 big squares apart ($d=6$) trace the equipotentials using the “pencil” tool. Trace one equipotential along the line connecting the charge every big square. Show that for about $d=4/6$ from red $\rightarrow V=0$ Then Keep tracing and show that there is another place where $V=0$ for $2d$ (12 squares from red

Then to get the big picture, place them 2 squares apart. See whats going on far from the blue Charge (right). Trace on a paper the voltage as a function of distance along the axis.



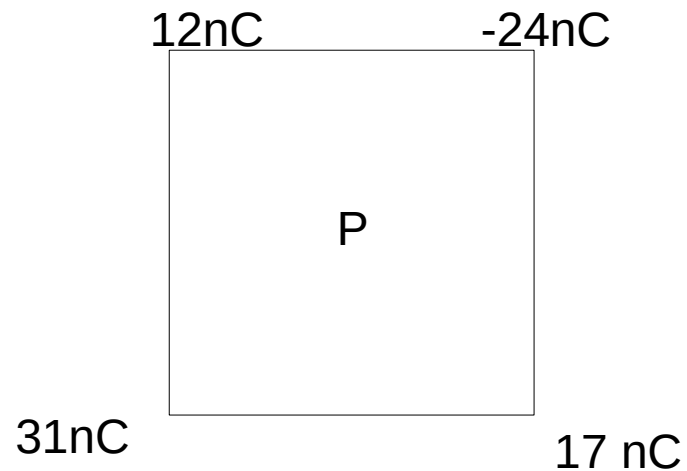
popquiz



Conceptual Example 7 Where is the Potential Zero?

Two point charges are fixed in place. The positive charge is $+2q$ and the negative charge is $-q$. On the line that passes through the charges, how many places are there at which the total potential is zero?

You can use your algebraic skills and show that you can find 2 points where $V = 0$
Or better. Use the simulation to visualize the contour lines.
See next slides

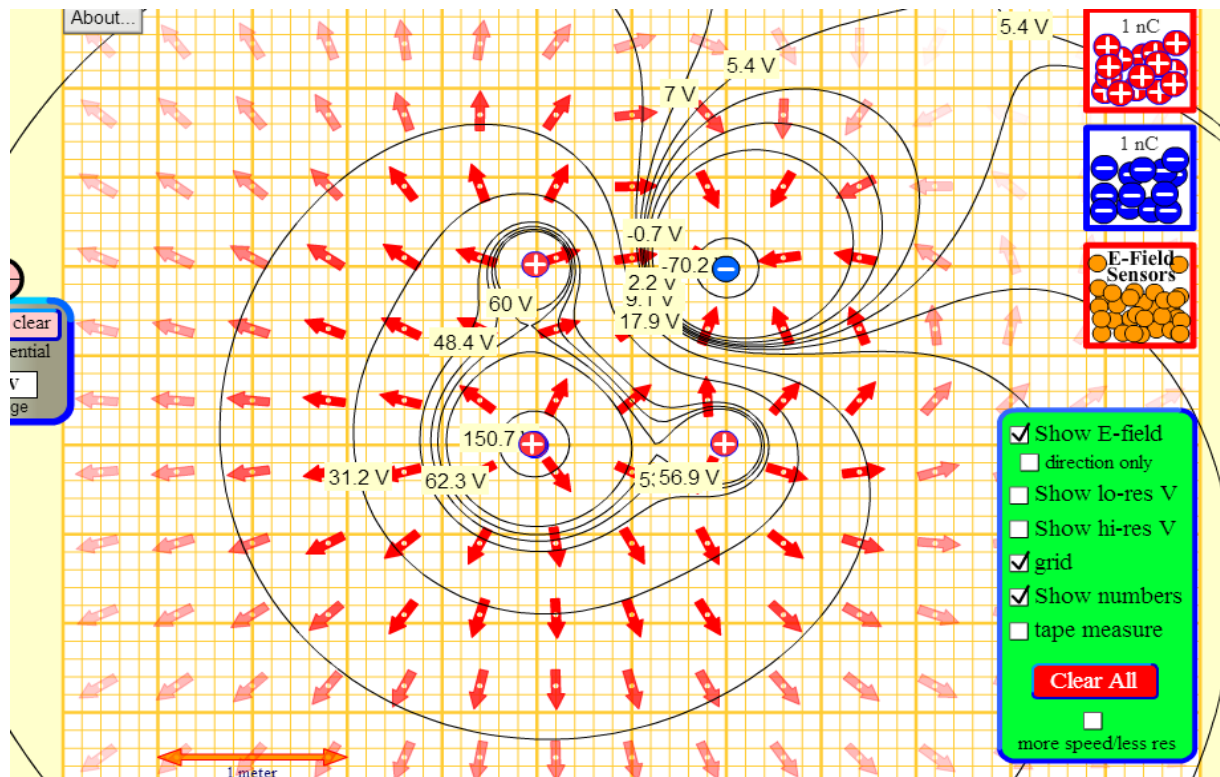


Potential at a given point due
To a distribution of charged.
Just

$V = \text{Sum}(Kq / r)$ with q can be
Negative. Its a scalar.

**We can fist trace the equipotential
Then trace the electric field lines !**

<https://phet.colorado.edu/en/simulation/charges-and-fields>

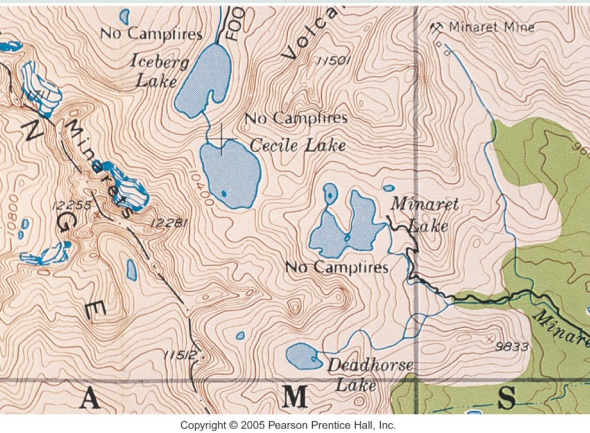


You see that the electric field
Is always perpendicular
To the equipotentials.

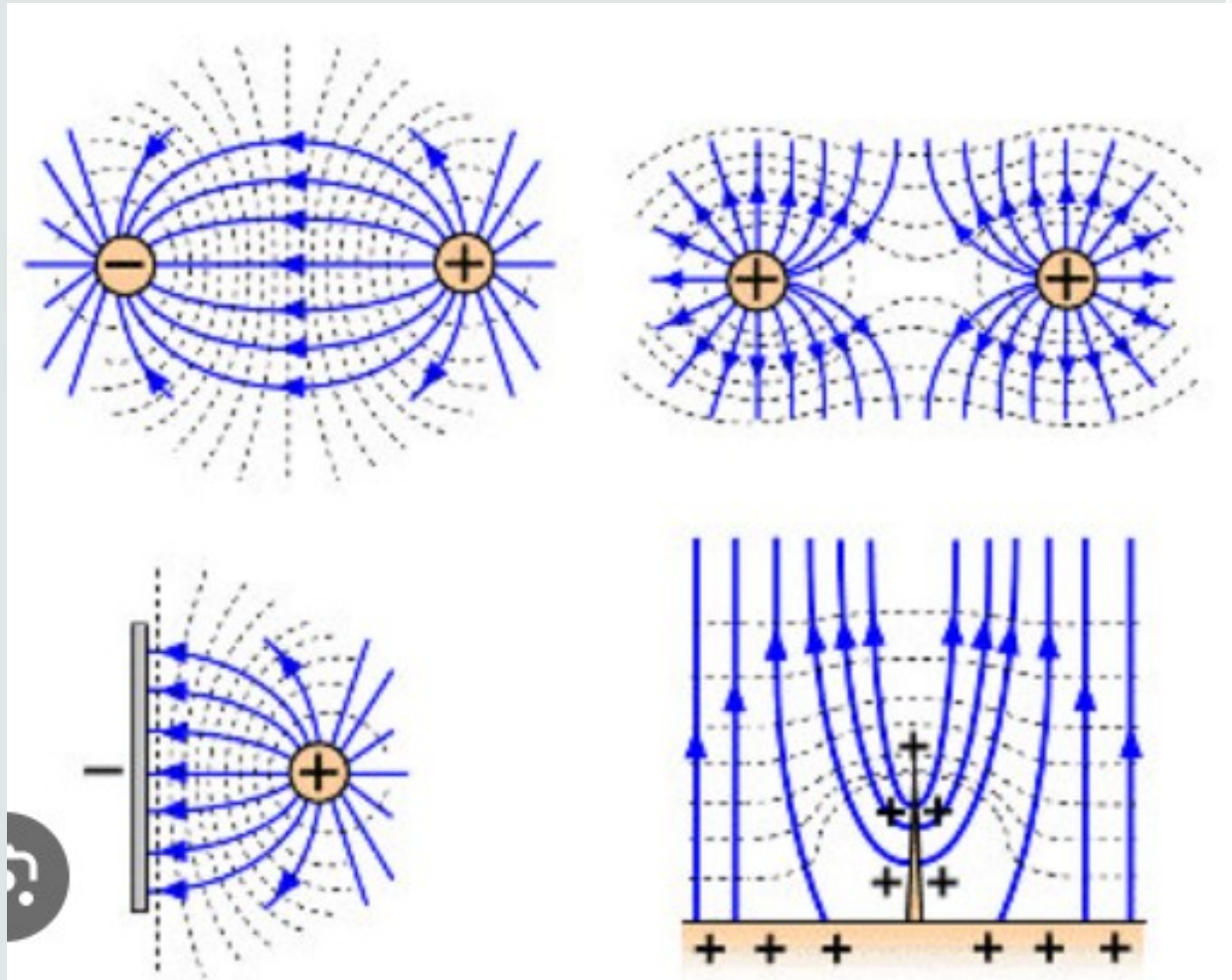
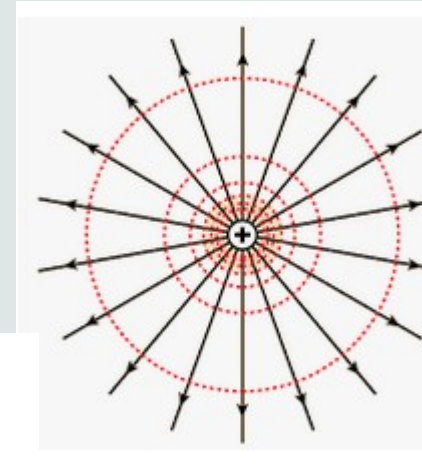
The arrow show the direction
Of the steepest slope.

The electric field is
 $E = - \text{grad}(V)$
→ greatest slope

Also. No work is done
If a charge is move along
equipotential



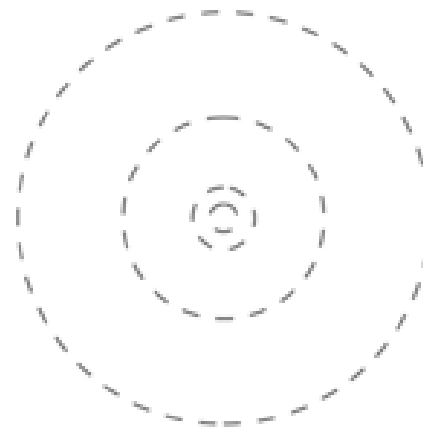
Electric field is always perpendicular to equipotential



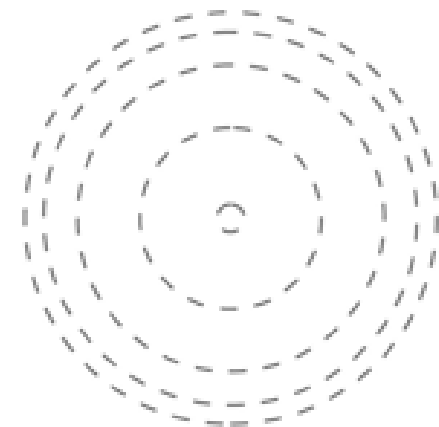
No work is done along the contour lines.

- The figure shows cross sections through two equipotential surfaces. In both diagrams the potential difference between adjacent equipotentials is the same. Which of these two could represent the field of a point charge?

- A. (a)
- B. (b)
- C. neither (a) nor (b)



(a)



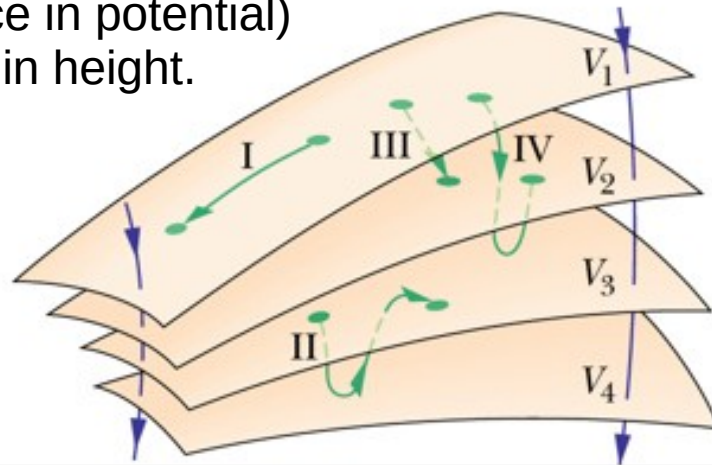
(b)

Work: positive or negative?

3. The right figure shows a family of equipotential surfaces associated with the electric field due to some distribution of charges. $V_1=100$ V, $V_2=80$ V, $V_3=60$ V, $V_4=40$ V. W_I , W_{II} , W_{III} and W_{IV} are the works done by the electric field on a charged particle q as the particle moves from one end to the other. Which statement of the following is not true?

Work done by $E = -$ (difference in potential)
Like difference in height.

- A. $W_I = W_{II}$
- B. W_{III} is not equal to zero
- C. W_{II} equals to zero
- D. $W_{III} = W_{IV}$
- E. W_{IV} is positive



From the Heart...

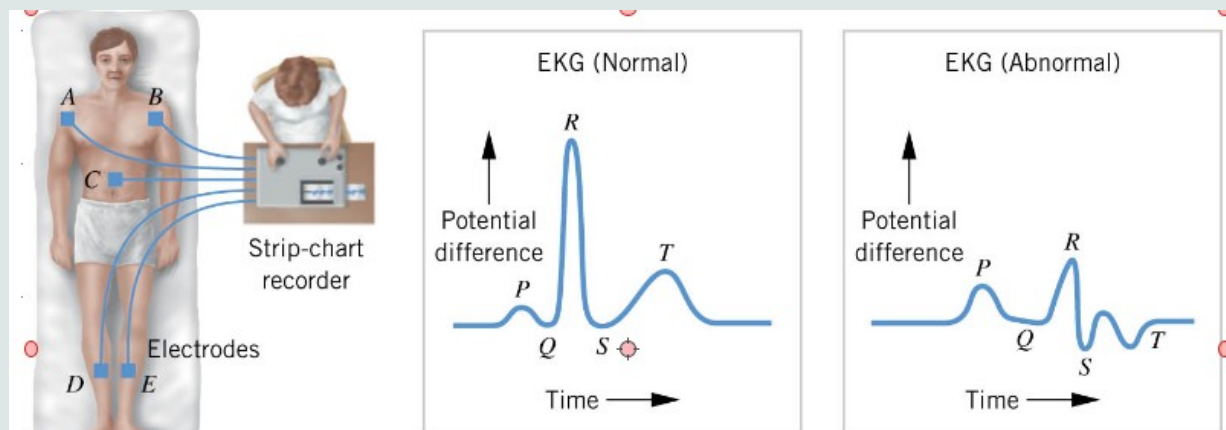
Negative
potential

Positive
potential



EKG

Voltage differences no greater than 1mV
Between different part of the skin.



The electric field points in the direction in which the electric potential most rapidly decreases.
The electric field intensity at a point is the gradient of the electric potential at that point
“Electric field is the negative space derivative of electric potential.”

Calculating the Field from the Potential

- Suppose that a positive test charge q_0 moves through a displacement ds from on equipotential surface to the adjacent surface.
- The work done by the electric field on the test charge is $W = -dU = -q_0 dV$.
- The work done by the electric field may also be written as $W = q_0 \vec{E} \cdot d\vec{s}$

- Then, we have

$$-q_0 dV = q_0 E (\cos \theta) ds$$

$$E \cos \theta = -\frac{dV}{ds}$$

- So, the component of E in any direction is the negative of the rate at which the electric potential changes with distance in that direction.

$$E_s = -\frac{\partial V}{\partial s}$$

- If we know $V(x, y, z)$,

$$E_x = -\frac{\partial V}{\partial x}$$

$$E_y = -\frac{\partial V}{\partial y}$$

$$E_z = -\frac{\partial V}{\partial z}$$

The magnitude of the electric field at any point is equal to the magnitude of the maximum rate of change of voltage with distance at that point. In mathematical terms
The direction of the electric field is the direction in which the voltage decreases most rapidly:

Major Concepts:

E and V

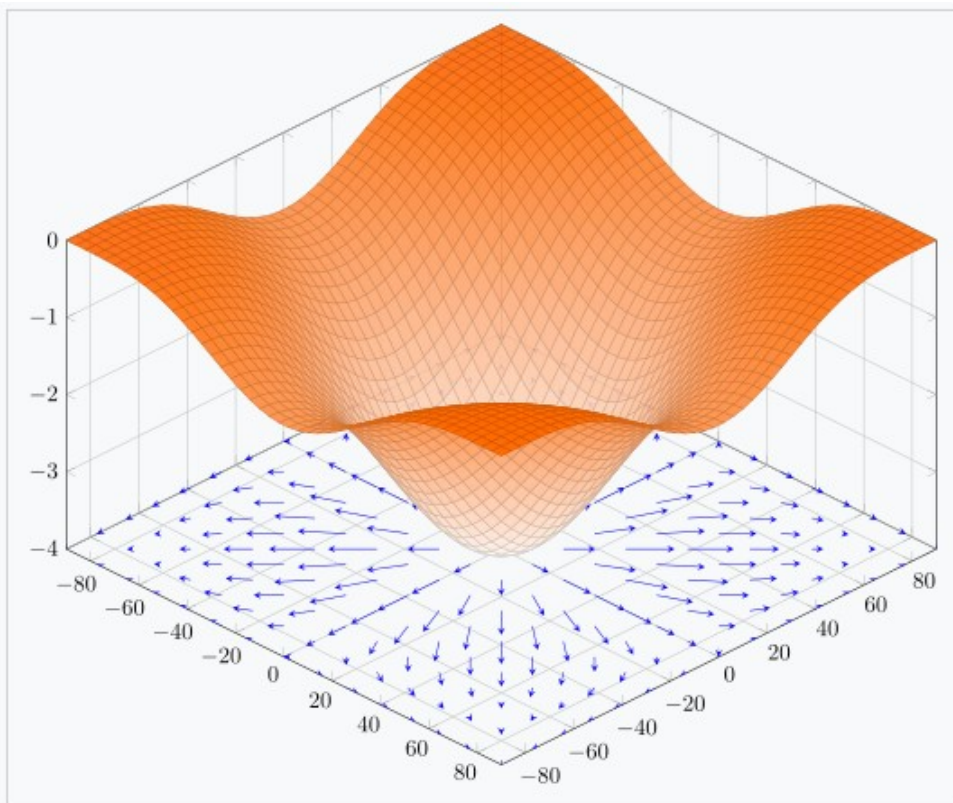
$$\bar{E} = -\nabla V$$



$$V_{BA} = V_B - V_A = -\int_A^B \bar{E} \cdot d\bar{\ell}$$

$$E = iE_x + jE_y + kE_z = -i\frac{\partial V}{\partial x} - j\frac{\partial V}{\partial y} - k\frac{\partial V}{\partial z}$$

$$= -\left[i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z} \right] V$$



The gradient of the function $f(x,y) = -(\cos^2 x + \cos^2 y)^2$ depicted as a projected **vector field** on the bottom plane.

Use wolfram alpha

Plot x^2+y^2

And plot $\text{gradient}(X^2+y^2)$

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k},$$

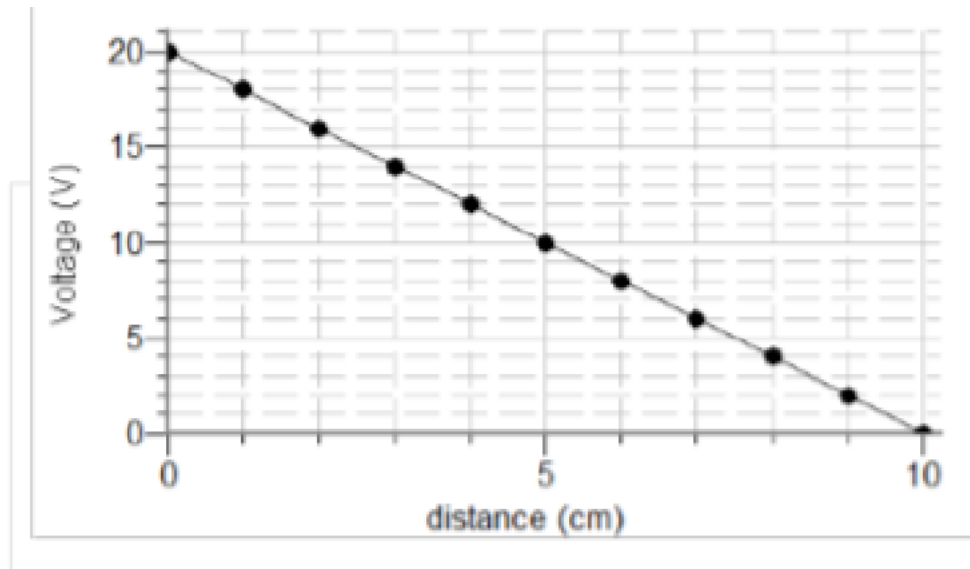
where $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are the **standard** unit vectors in the directions of the x , y and z coordinates, respectively. For example, the gradient of the function

$$f(x, y, z) = 2x + 3y^2 - \sin(z)$$

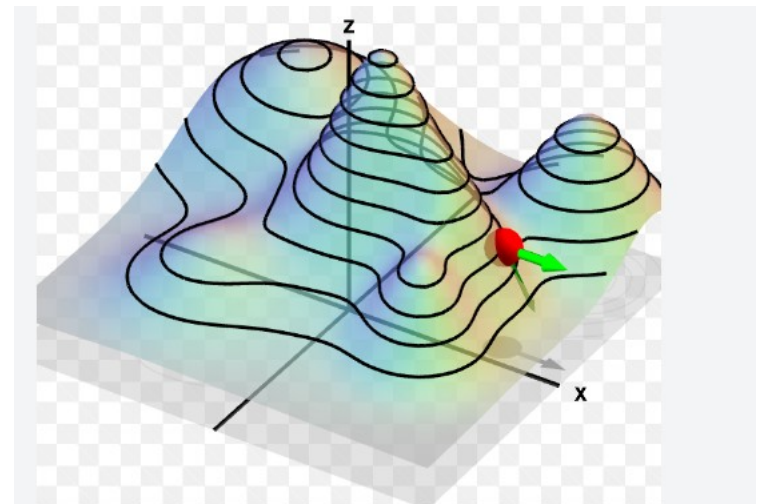
is

$$\nabla f(x, y, z) = 2\mathbf{i} + 6y\mathbf{j} - \cos(z)\mathbf{k}.$$

How do you find the steepest slope of a function? Take its derivative and look for the largest and smallest (most negative) value you can get out of out that function. The derivative is the gradient of the curve. Aug 25, 2017

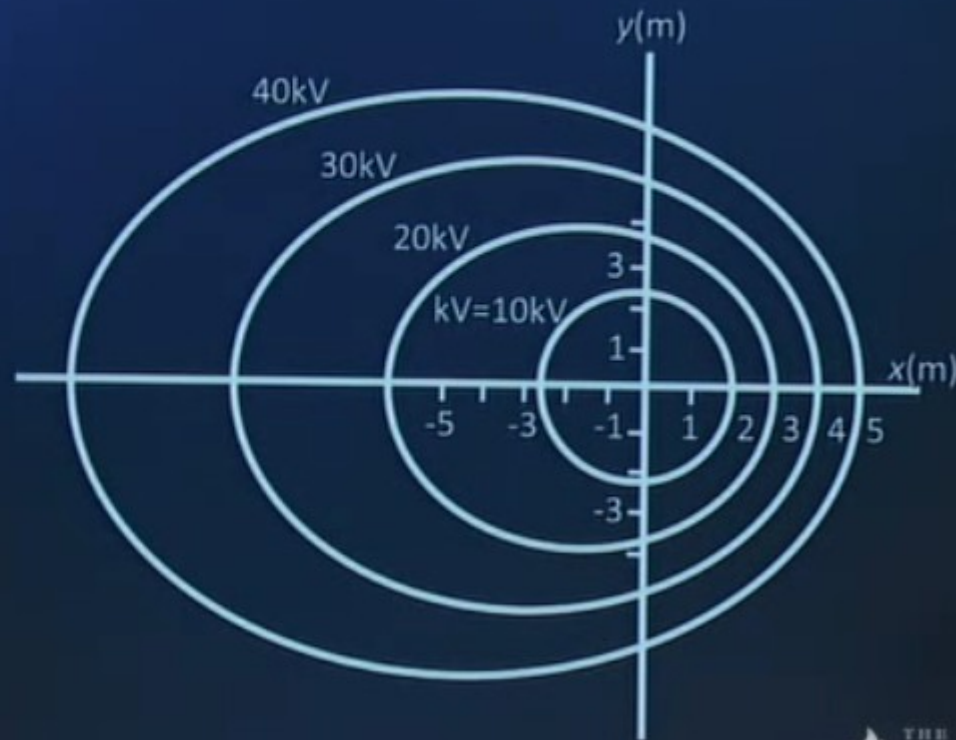


Potential between the plate of a capacitor.
What is E at 4cm? From the positive plate?



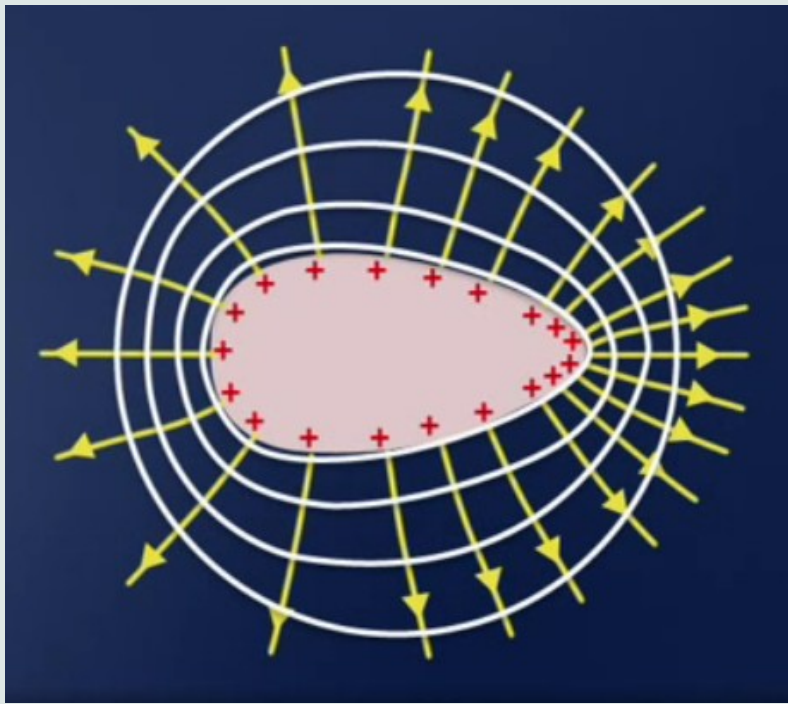
Using Equipotentials

What's the greatest electric field here?



What's the greatest electric field on the map?
What is the force on a 1MC force?

E is the slope of the equipotentials. Showing which way to go to decrease the potential
The fastest. $E_x = - \Delta V / \Delta X$



System designed to avoid sharp Corners → corona discharge

V at the surface of a spherical conductor = $q K / R$
(radius of conductor). Inside $V = \text{constant}$.

E at the surface = $q K / R^2$

So $V = R E$.

For discharge to happen → $E_{\text{max}} = 3,000,000 \text{ N/C}$

There is a breakdown of the electric field and the conductor

Discharges. So $V_{\text{max}} = R E_{\text{max}}$

So larger R → large voltage we can build up

Van Graaf generator if $R = 1\text{m}$ then $V = 3 \text{ million V}$

In class **$R = 20\text{cm}$ → $600,000 \text{ V}$**

(but low energy, not many charges)



Lightning rods have large
Curvature → can take more
Charge build up, more voltage.

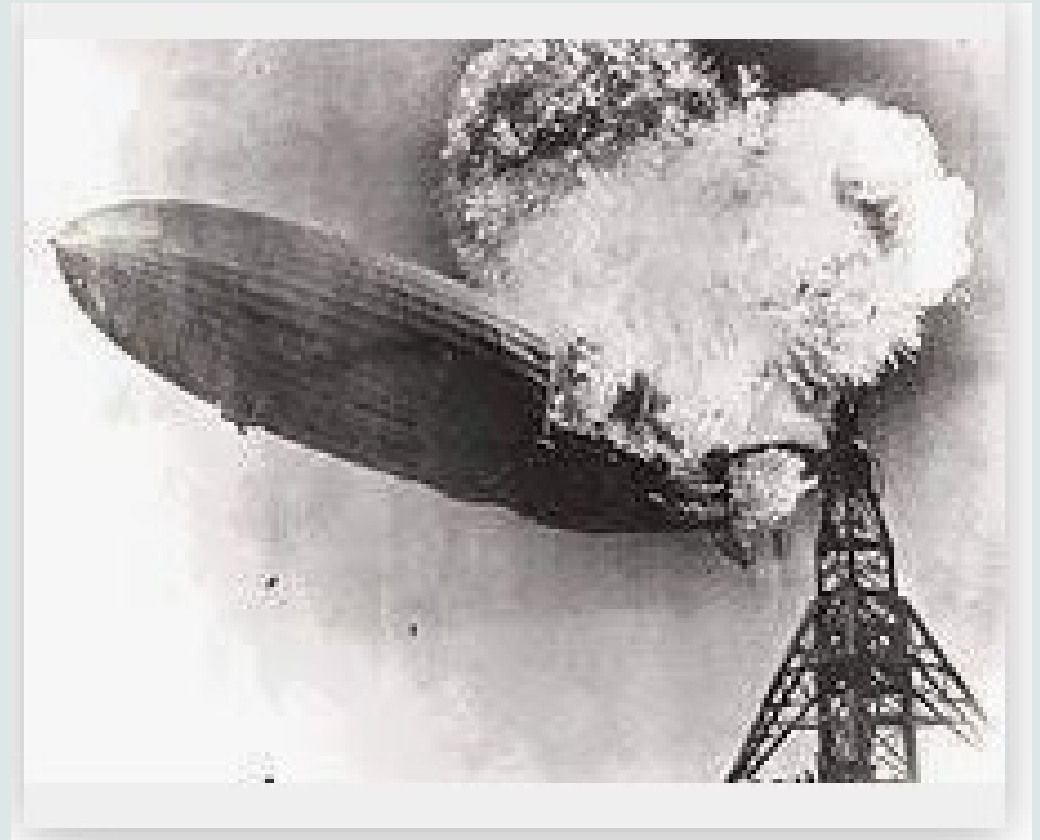
Then charge are grounded.



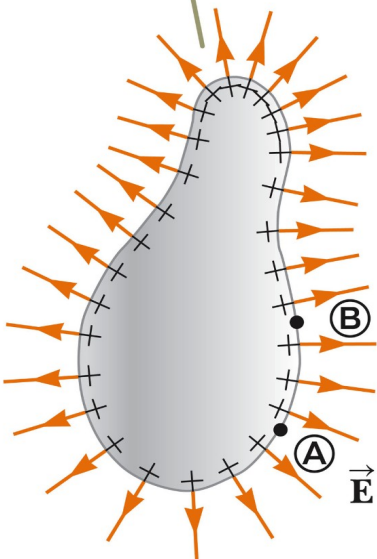


Hindenburg disaster. Corona discharge.

<https://www.youtube.com/watch?v=F54rqDh2mWA>



Notice from the spacing of the positive signs that the surface charge density is nonuniform.



E is the force per unit charge (1C) in magnitude

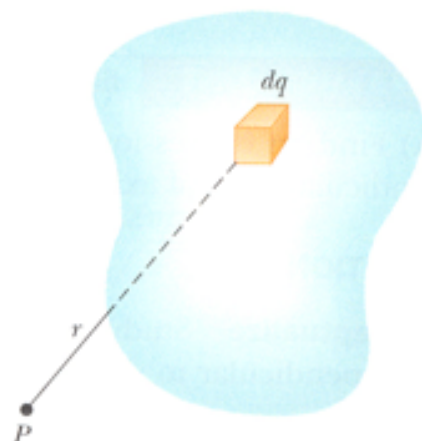
V is the potential energy per unit charge (1C)

E derives from V

Restoring forces (spring force, gravity) derives

From potential energy

Potential due to a Continuous Charge Distribution

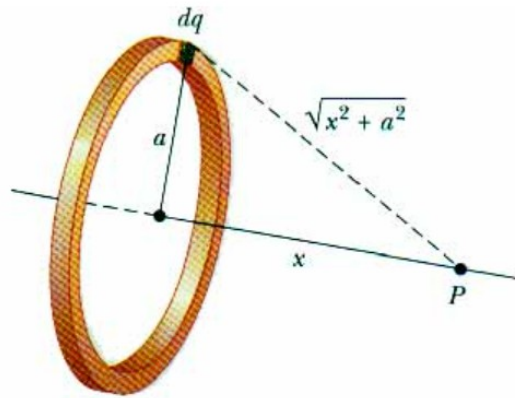


- Find an expression for dq :
 - $dq = \lambda dl$ for a line distribution
 - $dq = \sigma dA$ for a surface distribution
 - $dq = \rho dV$ for a volume distribution
- Represent field contributions at P due to point charges dq located in the distribution.

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

- Integrate the contributions over the whole distribution, varying the displacement as needed,

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$



Consider a **ring** of charge. The ring has radius a and we want to calculate the electric potential at a point P on the axis of the ring, a distance x away from the ring.

$$V = k \int \frac{dq}{r} = k \int \frac{dq}{\sqrt{x^2 + a^2}}$$

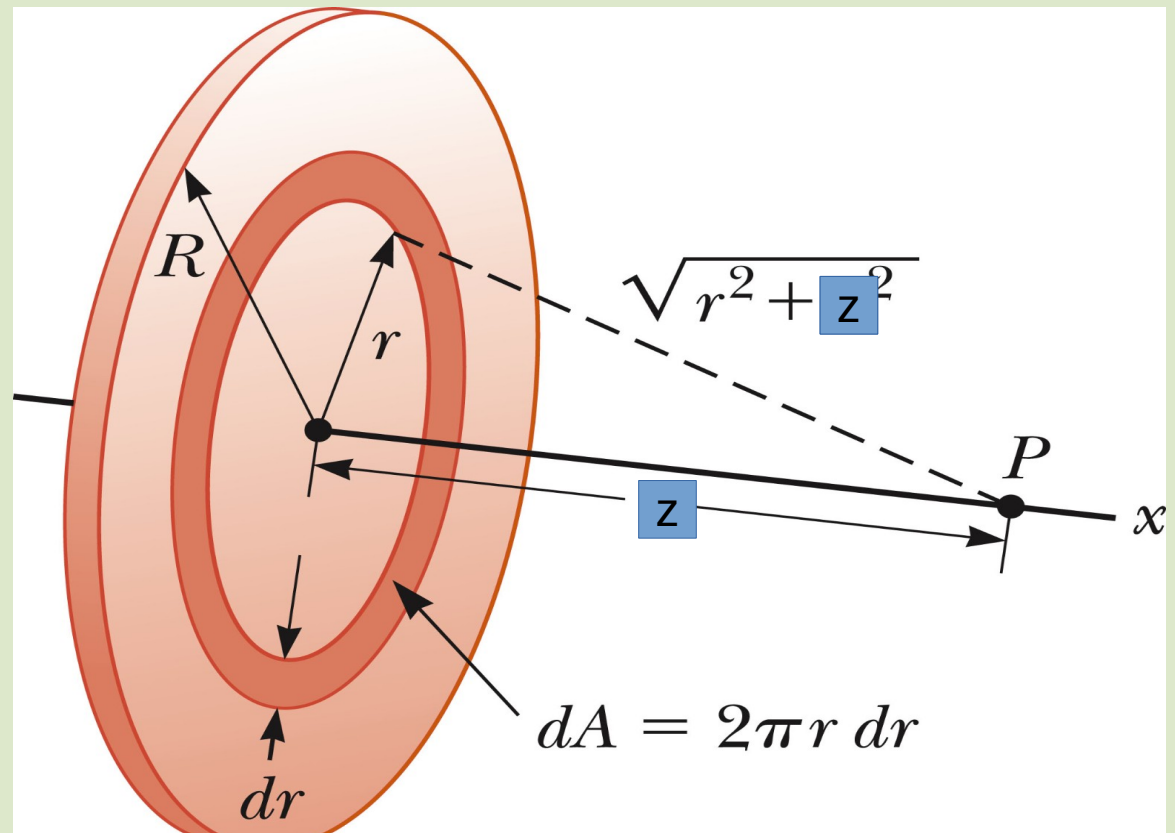
$$V = \frac{k}{\sqrt{x^2 + a^2}} \int dq$$

$$V = \frac{k}{\sqrt{x^2 + a^2}} Q$$

<http://hyperphysics.phy-astr.gsu.edu/hbase/electric/potlin.html#c1>

Textbook23.11

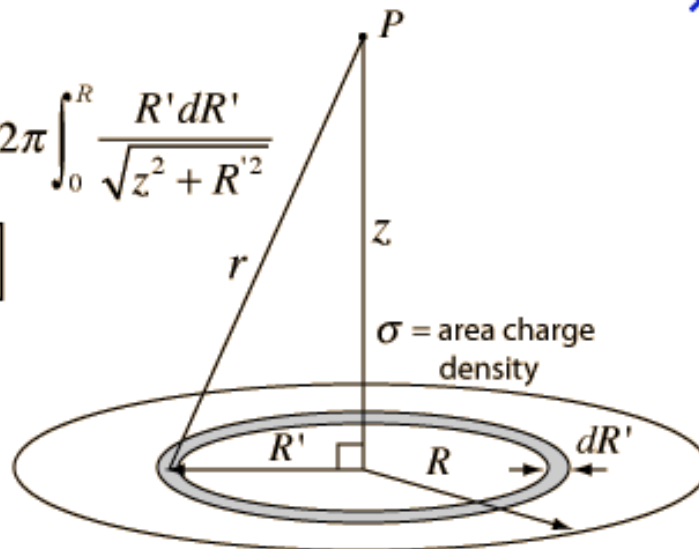
7. charged disk



$$V = k\sigma 2\pi \int_0^R \frac{R' dR'}{r} = k\sigma 2\pi \int_0^R \frac{R' dR'}{\sqrt{z^2 + R'^2}}$$

$$V = k\sigma 2\pi \left[\sqrt{z^2 + R^2} - z \right]$$

$$k = \frac{1}{4\pi\epsilon_0} = \text{Coulomb's constant}$$



×

Build up on previous problem

!!Note!!: $z=0$ $V=K \sigma 2\pi R$
Or $V = \sigma R / 2\epsilon_0$

Example: Potential Due to a Charged Rod

- A rod of length L located along the x axis has a uniform linear charge density λ . Find the electric potential at a point P located on the y axis a distance d from the origin.

- Start with

$$dq = \lambda dx$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x^2 + d^2)^{1/2}}$$

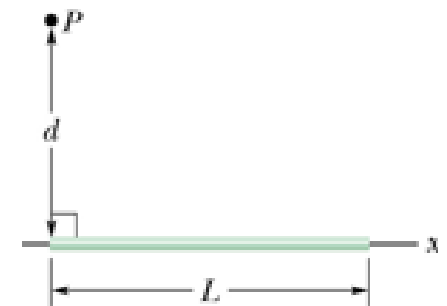
- then,

$$\begin{aligned} V &= \int dV = \int_0^L \frac{\lambda}{4\pi\epsilon_0} \frac{dx}{(x^2 + d^2)^{1/2}} = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + (x^2 + d^2)^{1/2}) \right]_0^L \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[\ln(L + (L^2 + d^2)^{1/2}) - \ln d \right] \end{aligned}$$

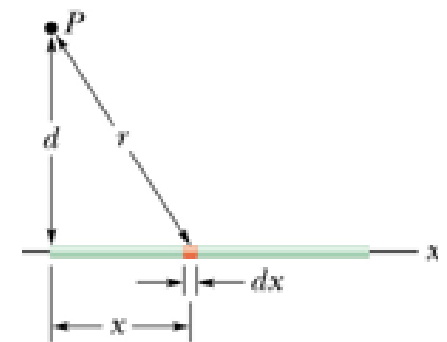
- So

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L + (L^2 + d^2)^{1/2}}{d} \right]$$

Textbook 23.10



(a)



(b)

Table 1 Fields and Potentials for Some Charge Configurations

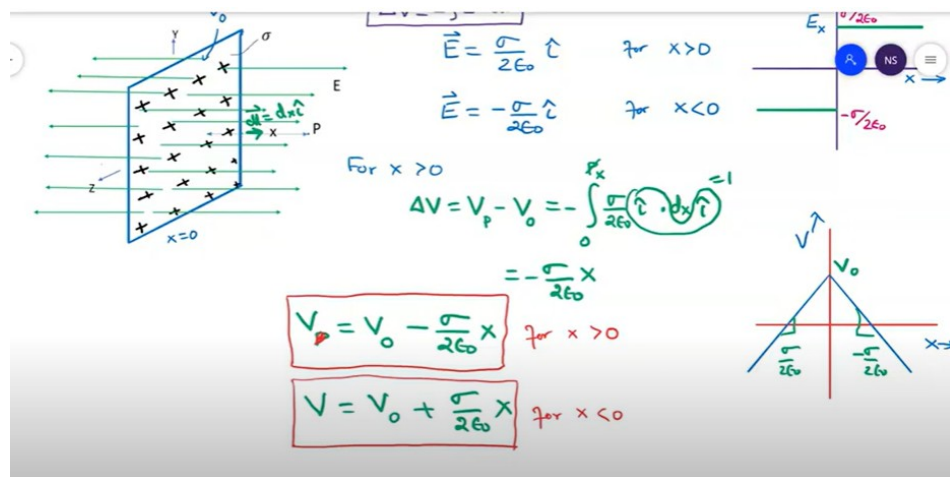
Configuration	Field E	Equation	Potential V	Equation
Point charge	$\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$	(24-6)	$\frac{1}{4\pi\epsilon_0} \frac{q}{r}$	(26-12)
Dipole ^a	$\frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$	(24-11)	$\frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$	(26-14)
Charged disk ^b	$\frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$	(24-21)	$\frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$	(26-17)
Infinite sheet	$\frac{\sigma}{2\epsilon_0}$	(25-15)	$V_0 - \left(\frac{\sigma}{2\epsilon_0} \right) z$	(Exercise 6)
Insulated conductor	$E = 0$, inside $E = \sigma/\epsilon_0$, at surface		$V = \text{a constant}$, inside and on the surface	

^a The field equation is for distant *axial* points. The potential equation is for *all* distant points.

^b For *axial* points.

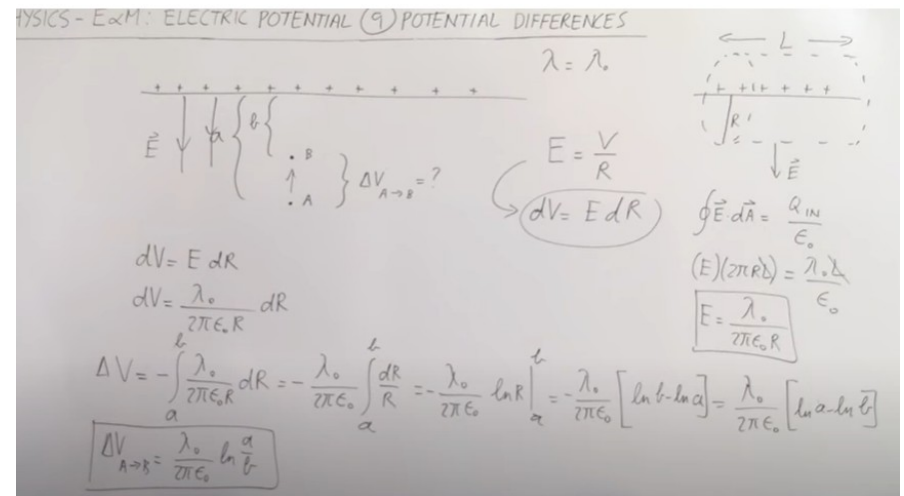
https://www.youtube.com/watch?v=jFL7jsTa_24

Nice tutorial for the infinite sheet →



https://www.youtube.com/watch?v=ujANn_-NNj8

Nice tutorial for the infinite charged Wire



Conservation of energy



We now include electric potential energy EPE as part of the total energy that an object can have:

$$E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + mgh + \frac{1}{2}kx^2 + EPE$$

- The change in potential energy is directly related to the change in voltage.

$$\Delta U = q\Delta V$$

$$\Delta V = \Delta U/q$$

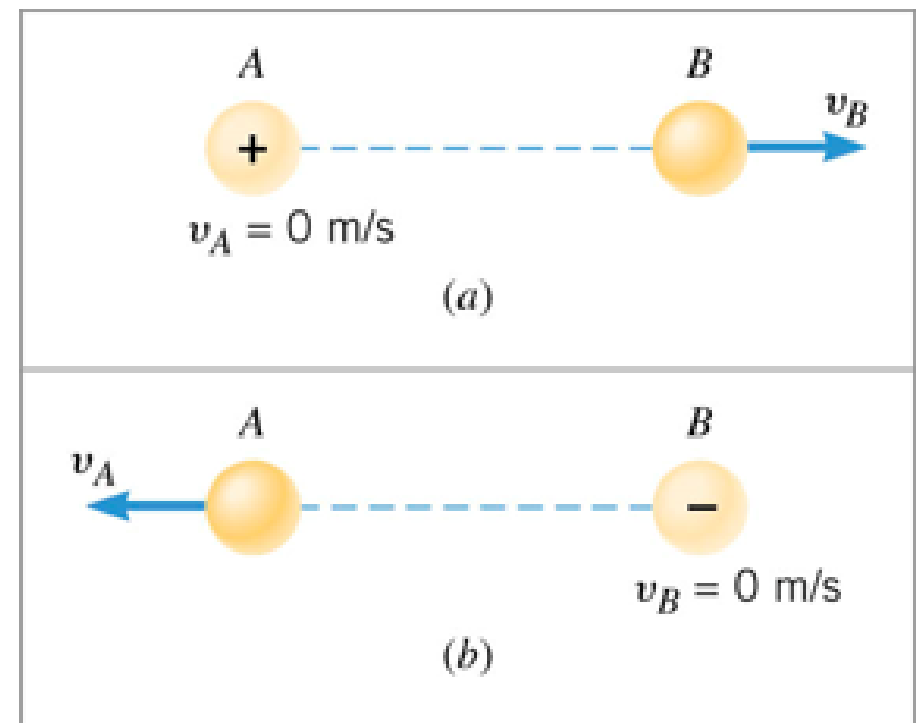
- ΔU : change in electrical potential energy (J)
- q : charge moved (C)
- ΔV : potential difference (V)
- All charges will spontaneously go to lower potential energies if they are allowed to move.

One electron volt is the magnitude of the amount by which the potential energy of an electron changes when the electron moves through a potential difference of one volt.

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ V}$$

Example 4 The Conservation of Energy

A particle has a mass of $1.8 \times 10^{-5} \text{ kg}$ and a charge of $+3.0 \times 10^{-5} \text{ C}$. It is released from point A and accelerates horizontally until it reaches point B. The only force acting on the particle is the electric force, and the electric potential at A is 25V greater than at B. (a) What is the speed of the particle at point B? (b) If the same particle had a negative charge and were released from point B, what would be its speed at A?



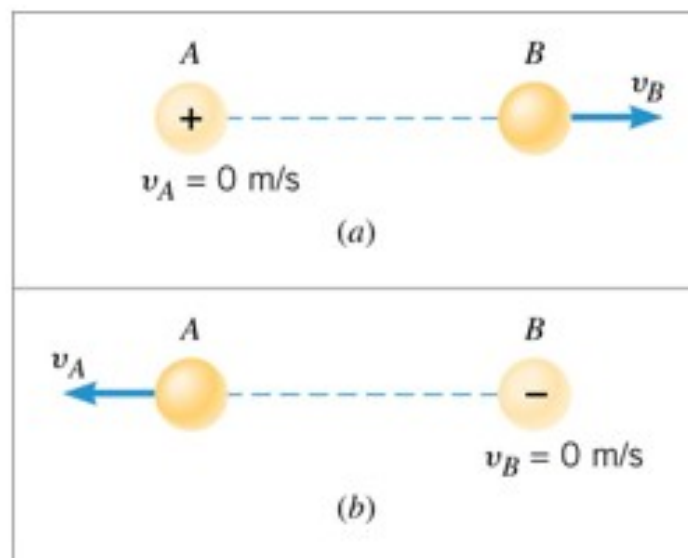
$$\frac{1}{2}mv_B^2 + EPE_B = \frac{1}{2}mv_A^2 + EPE_A$$



$$\frac{1}{2}mv_B^2 = \frac{1}{2}mv_A^2 + EPE_A - EPE_B$$



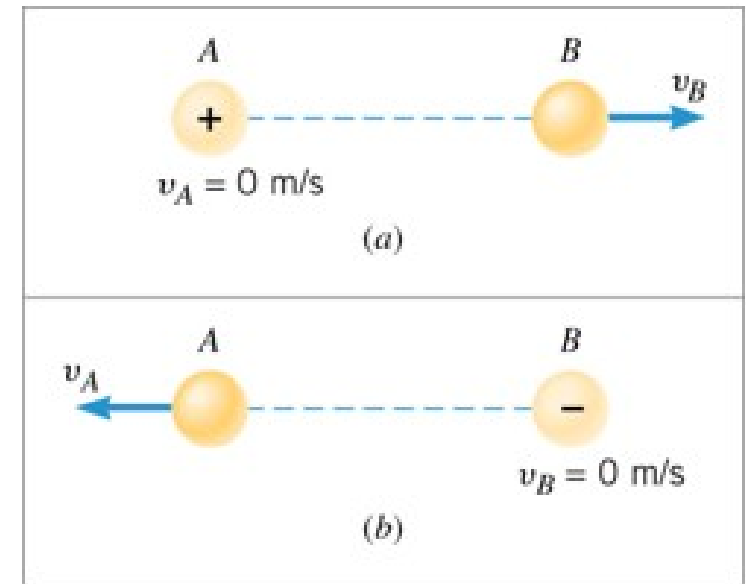
$$\frac{1}{2}mv_B^2 = \frac{1}{2}mv_A^2 + q_o(V_A - V_B)$$



$$(a) \quad \frac{1}{2}mv_B^2 = q_o(V_A - V_B)$$

$$v_B = \sqrt{2q_o(V_A - V_B)/m}$$

$$= \sqrt{2(3.0 \times 10^{-5} \text{ C})(25 \text{ V})/(1.8 \times 10^{-5} \text{ kg})} = 9.1 \text{ m/s}$$



$$(a) \quad v_A = \sqrt{-2q_o(V_A - V_B)/m}$$

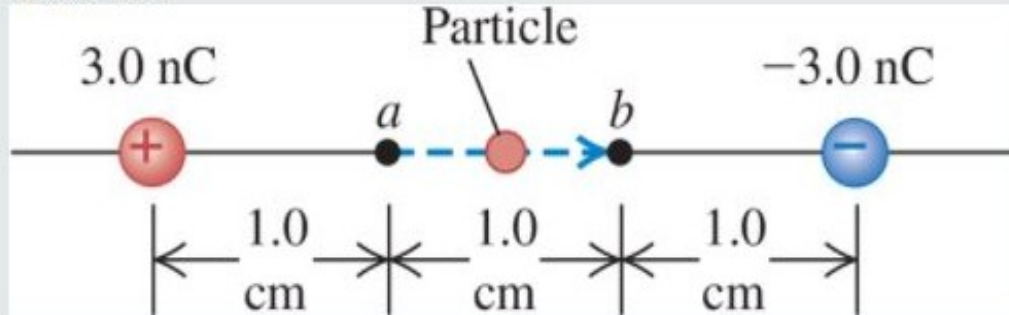
$$= \sqrt{-2(-3.0 \times 10^{-5} \text{ C})(25 \text{ V})/(1.8 \times 10^{-5} \text{ kg})} = 9.1 \text{ m/s}$$

A positive charge moves from high potential to low potential
A negative charge moves from low potential to high potential

23.7 Moving through a potential difference

In Fig. 23.15 a dust particle with mass $m = 5.0 \times 10^{-9} \text{ kg} = 5.0 \mu\text{g}$ and charge $q_0 = 2.0 \text{ nC}$ starts from rest and moves in a straight line from point a to point b . What is its speed v at point b ?

FIGURE 23.15



The particle moves from point a to point b ; its acceleration is not constant.

IDENTIFY and SET UP Only the conservative electric force acts on the particle, so the total mechanical energy is conserved: $K_a + U_a = K_b + U_b$. We get the potential energies U from the corresponding potentials V from Eq. (23.12): $U_a = q_0 V_a$ and $U_b = q_0 V_b$.

EXECUTE We have $K_a = 0$ and $K_b = \frac{1}{2} mv^2$. We substitute these and our expressions for U_a and U_b into the energy-conservation equation, then solve for v . We find

Textbook 23.7 next popquiz

A dust particle (charged q_0 and with a mass m) moves from a to b . What is speed At b .

- **First find V_a and V_b .**
- **find $V_a - V_b$**
- **$V_a - V_b = (U_{ab}) / q_0$ find the change in PE between a and b**
- **$U_{ab} = \text{change in KE}$**

Electrostatic Energy

- Point-charge potential

-Work *per unit charge* to move from infinity to distance r from point charge q_1 :

$$V = \frac{kq_1}{r}$$

- So work to bring in q_2

$$W = \frac{kq_1q_2}{r}$$

- Stored as potential energy

$$U = \frac{kq_1q_2}{r}$$



Or you can say. V is the potential energy to bring 1C of charge from infinity to position P (r)
So the energy to bring q_2 is $V q_2$. The potential is position because you had to do positive work
To bring it. If you release the charge, the PE goes to KE

Opposite Charges?

- Potential energy is still $U = \frac{kq_1q_2}{r}$
- Now it's *negative*
- U represents work required to separate the charges



The potential energy is negative. The charge is “sucked in”. Your work is negative/
The system is “glued”

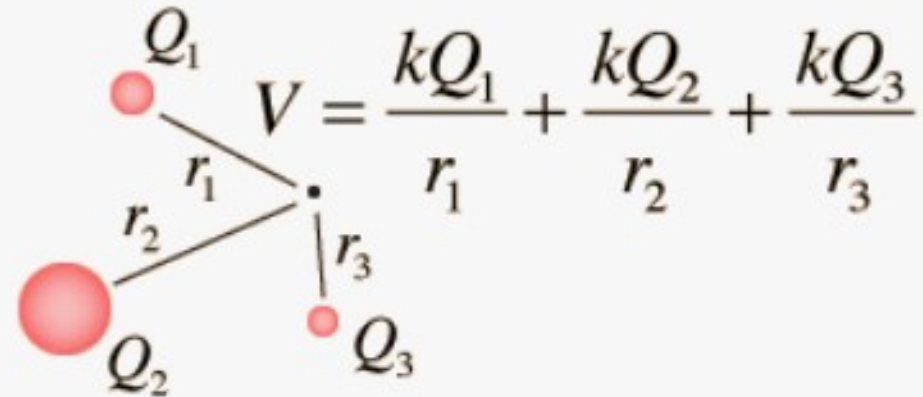
Electric potential energy versus voltage

$$U = \left[\frac{kq_1q_2}{r_{12}} + \frac{kq_2q_3}{r_{23}} + \frac{kq_1q_3}{r_{13}} \right]$$

Rearrange the expression as,

$$U = k \left[\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}} \right]$$

U is how much work was done to bring those charges together, It does not depend on the location in space.



V is the potential energy at point P relative to infinity of 1C of charge. qV is the potential energy of a charge q placed at P (like mgh for a mass m) The potential depends on the location P in space (like how high above ground)

POTENTIAL DIFFERENCE AND ELECTRICAL POTENTIAL ENERGY

The relationship between potential difference (or voltage) and electrical potential energy is given by

$$\Delta V = \frac{\Delta PE}{q}$$

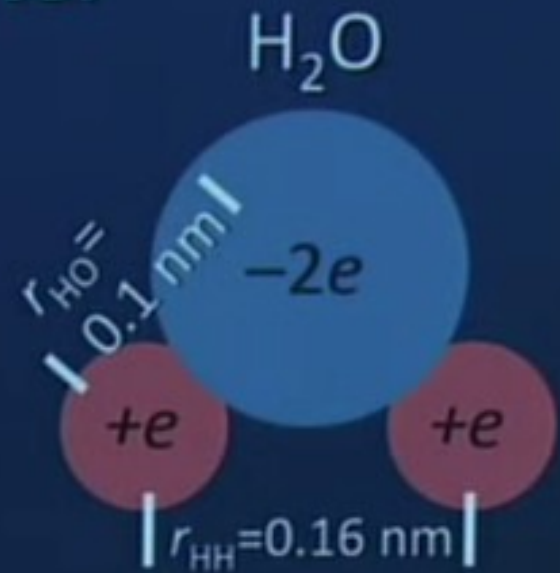
So if you need to find the PE of a charge q relative to a reference level, then multiply the charge by the voltage.

$$\text{and } \Delta PE = q\Delta V$$

Making Water

How potential energy is locked inside
A water molecule ?
Is it negative (binding energy) or positive?

Just apply the equation below



$$U = k \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right]$$

Making Water

Energy of 2 point charges: kq_1q_2/r

Start with oxygen:

Work to bring in 1st hydrogen:

$$W_1 = k(-2e)(e)/r_{HO} = -2ke^2/r_{HO}$$

Work to bring in 2nd hydrogen:

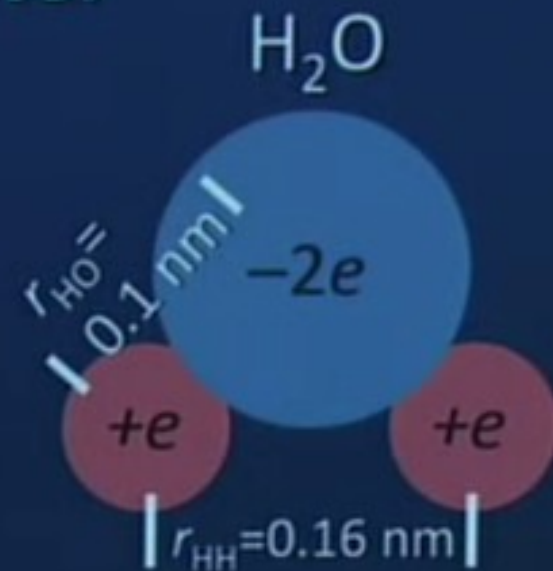
$$W_2 = -2ke^2/r_{HO} + ke^2/r_{HH}$$

Total energy in water molecule:

$$U = W_1 + W_2 = -4ke^2/r_{HO} + ke^2/r_{HH} = ke^2 \left(\frac{1}{r_{HH}} - \frac{4}{r_{HO}} \right)$$

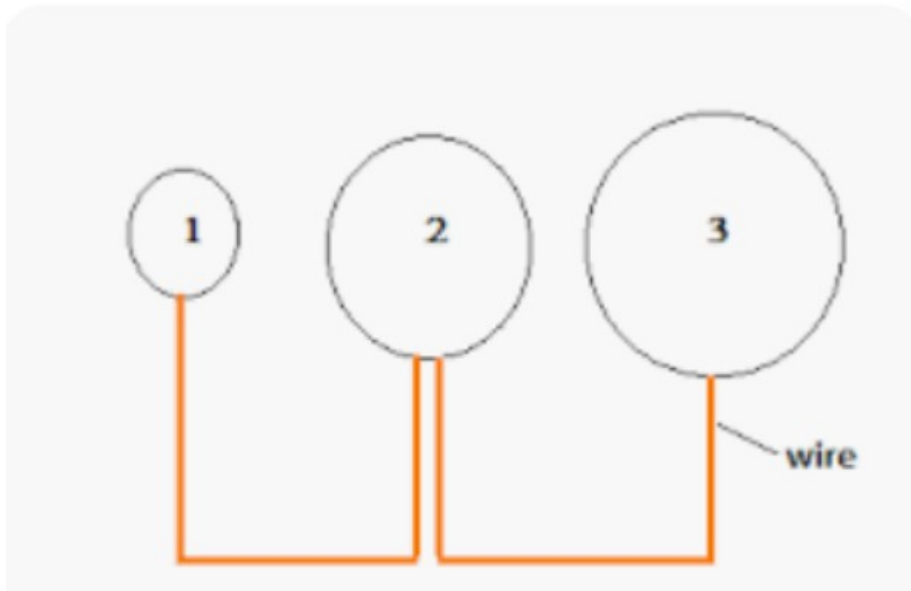
Do the numbers: $U =$

$$9 \cdot 10^{18} (e^2) \times (-4/0.1 + 1/0.16) = -8 \cdot 10^{-18} \text{ J}$$



$$U = k \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right]$$

Its negative so binding



3 conductive spheres are connected with a copper wire.

What is True?

E_1, E_2, E_3 are the electric fields at the surface of the spheres 1,2,3.

V_1, V_2, V_3 are the voltage of the spheres.

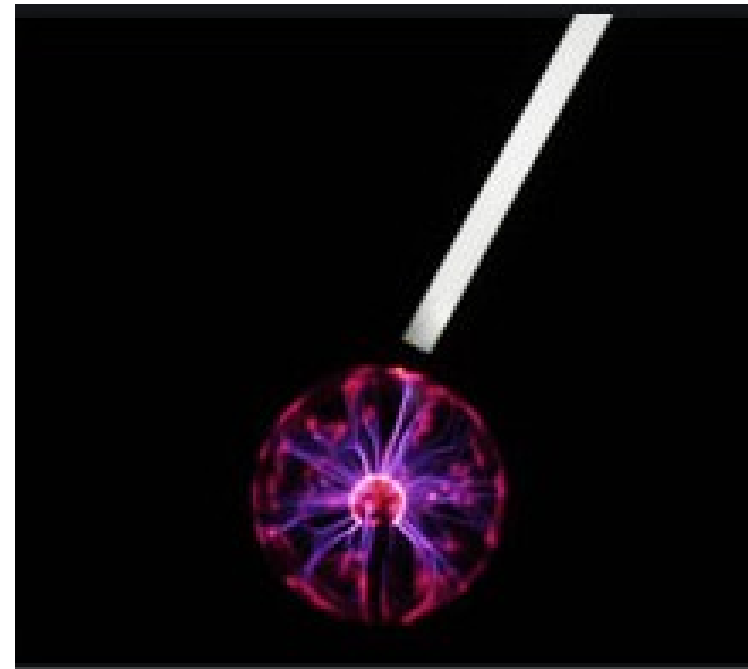
The 3 spheres are charged (one was charged with a Van Graaf then the connection was done carefully)

Those 3 conductive spheres are connected to each other with a conductive wire. One sphere is charged with a Van graaf generator.

What happens next ? are the charges spread or stay on the sphere connected to Van Graaf ? What about the potential V_1, V_2, V_3 ? Discuss

And what about the electric field ? Hint: the sphere 1 has a smaller curvature.

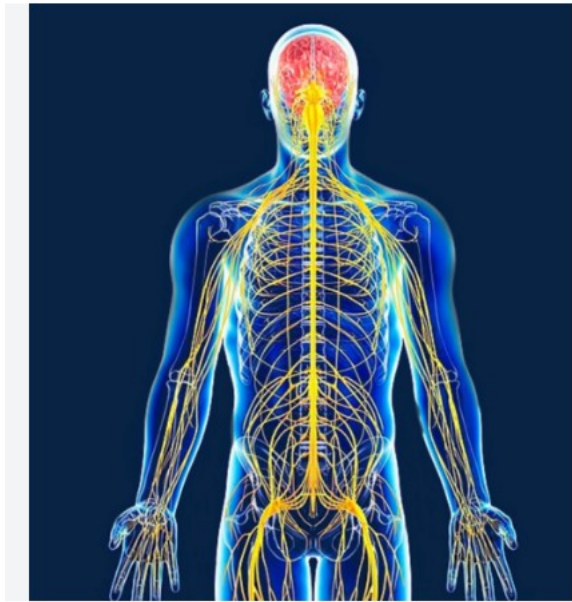
Take a guess. Then do the Math.



Go to 4:04

Demo in
class

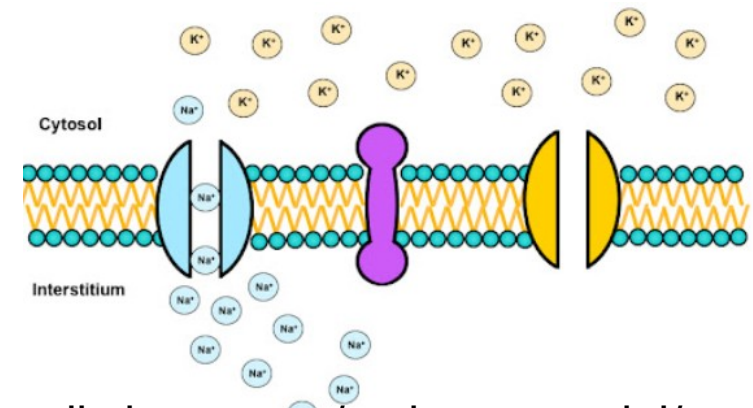
<https://www.youtube.com/watch?v=VT4zRPIU-3E>



Signal transmission along neurons does not happen because of a flow of charge carriers but transmission relies on charges moving in and out of the cell membranes → It's a change in voltage that is propagating.

<https://www.livescience.com/27975-human-body-system-the-nervous-system-infographic.html>

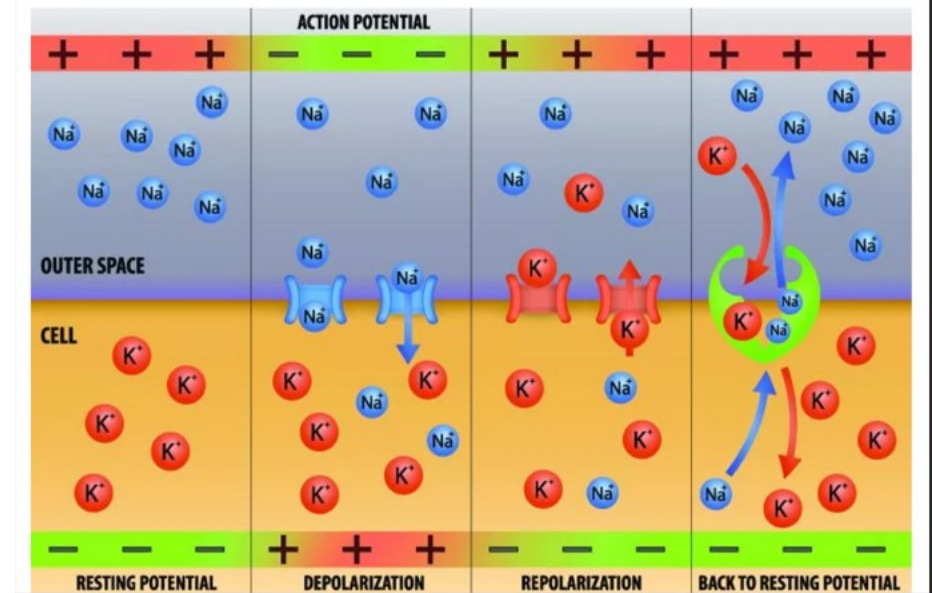
The interior of a human cell is at a lower electric potential V than the exterior.
 - 70mV in neurons and -95mV for muscle cells.
 E is from outside to inside → control the flow of ions.

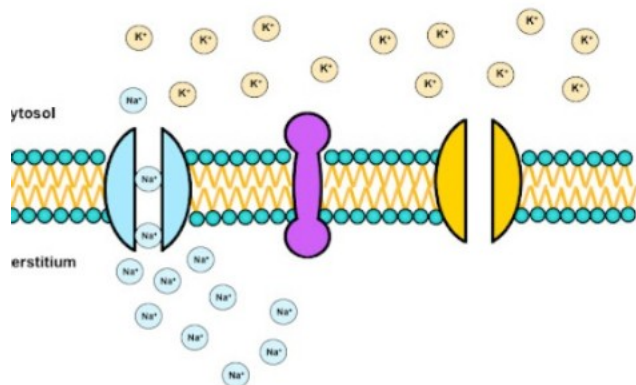


<https://biologydictionary.net/action-potential/>

An action potential is the result of a very rapid rise and fall in voltage across a

cellular membrane, with every action potential (impulse) similar in size. The response of a nerve or muscle cell to an action potential can vary according to how frequently and for what duration the action potentials are fired. An action potential requires an influx of positive ions to produce a specific change in voltage (threshold value). It occurs after a certain degree of internal cell membrane depolarization – a positive increase in electrical charge.





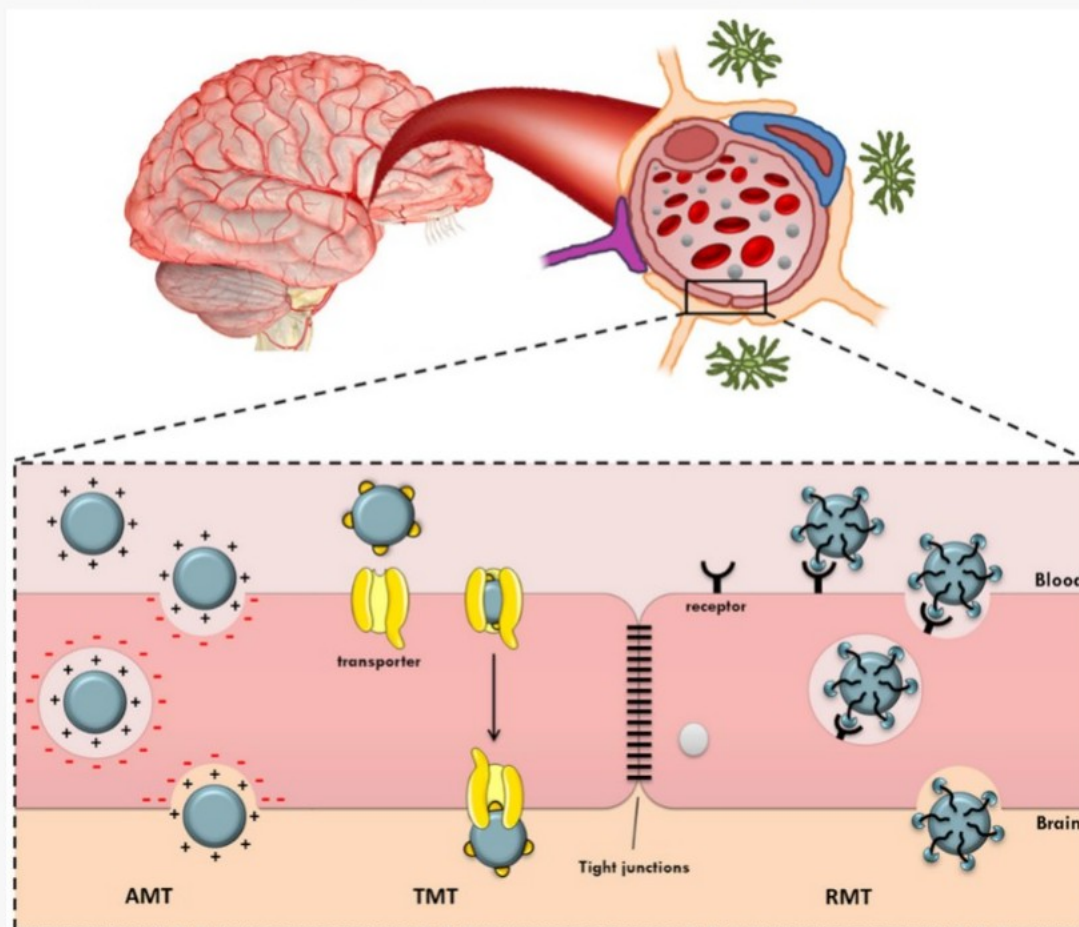
The interior of a human cell is at a lower electric potential V than the exterior.

To deliver drugs to the brain,
The barrier blood/brain
Has to be crossed.

They use gene therapy →
Lipid nanoparticles
That
Are charged and can get to
The cell of the brain.

For serious disease (risk and benefit balance)

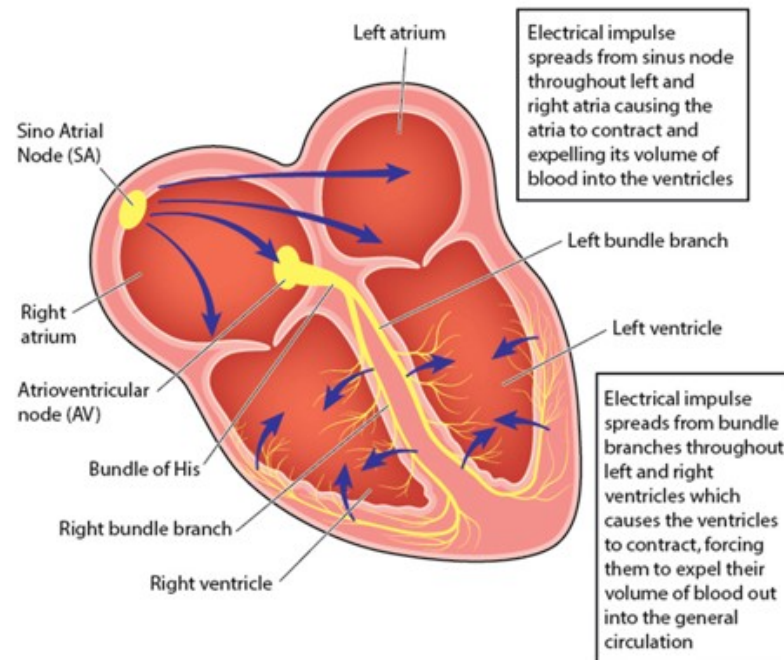
Figure 1. Schematic representation of the different mechanisms of nanoparticles for crossing the blood–brain barrier (BBB). AMT, adsorptive-mediated transcytosis; TMT, transporter-mediated transcytosis; RMT, receptor-mediated transcytosis.

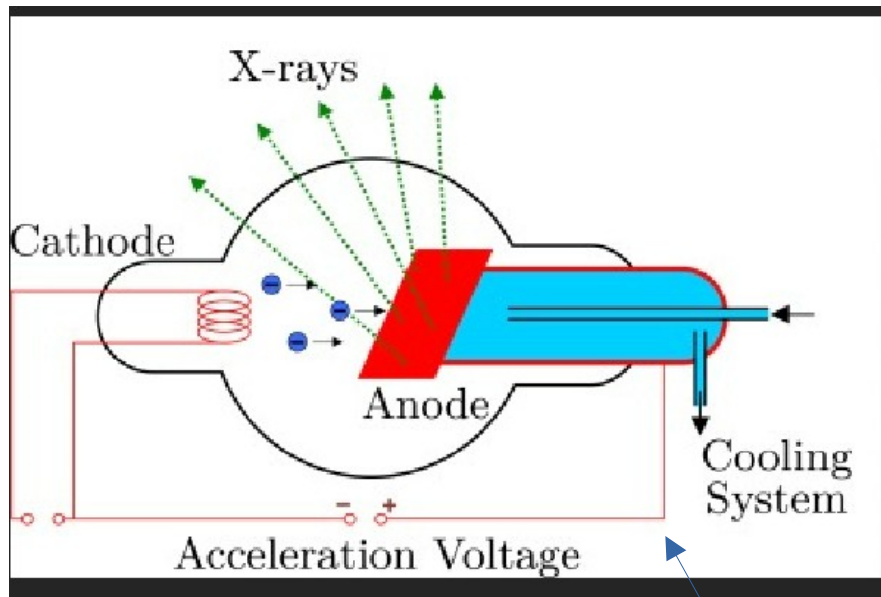


Likewise, in the heart. Signal propagates because charges move in and out of cells. But in the heart, Ca^{++} are the charge carrier.

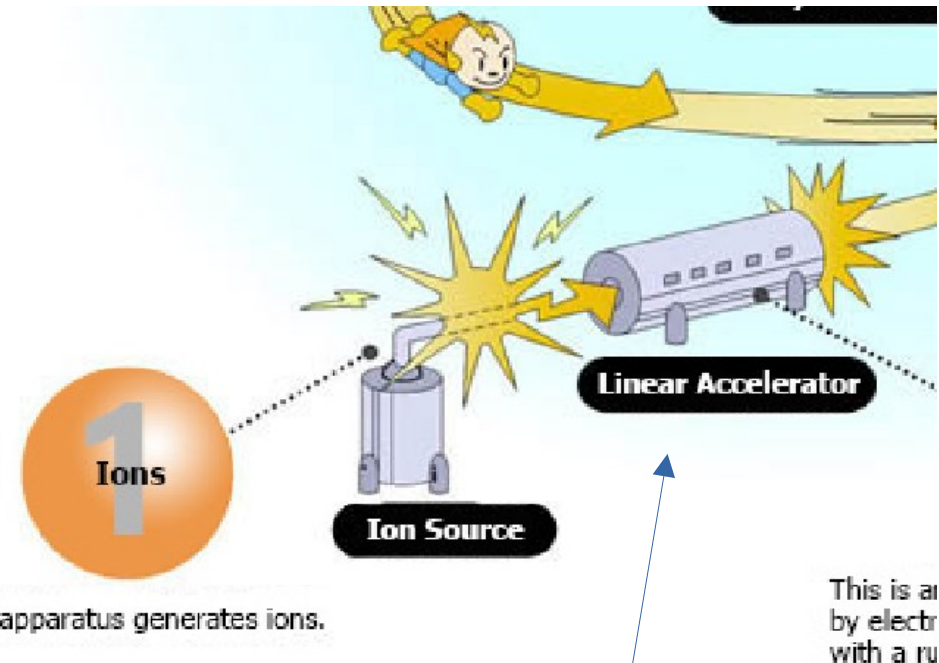
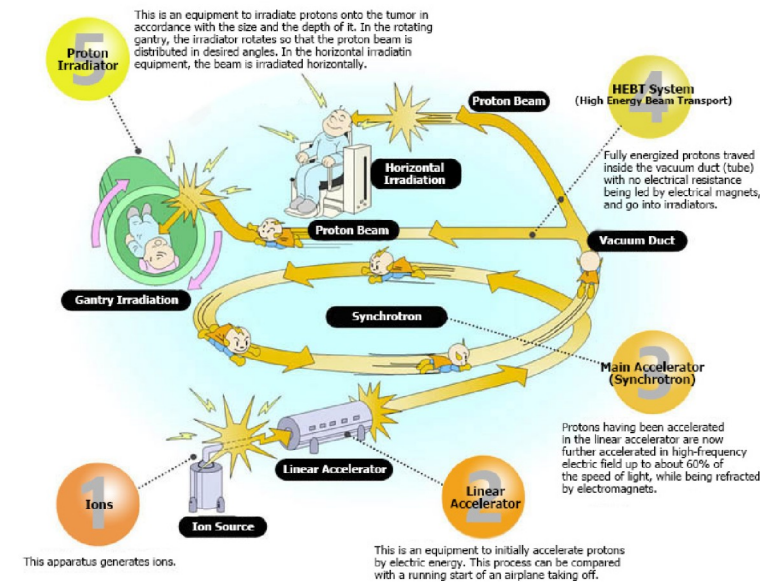
The heart conduction system(cardiac electrical system) controls heartbeats. If inflammation → Myocarditis. Cardiac arrest?

Heart Conduction System





A voltage of 50 kVp will produce a spectrum of X-ray energies with the theoretical maximum being 50 keV. Show how the electric field is from The Anode to the Cathode. Electrons move against the electric field and accelerate. Their PE is voltage x charge. The charge is e ($1.6 \times 10^{-19} \text{C}$) So PE is in eV.



250 million electron volts (MeV). This means that the proton must pass between two plates that have a voltage of 250 million volts (MV) between the two of them.

To separate charge → work is done by external agent (like battery) → voltage builds